

Base-Rate Neglect in Description and Experience*

Jilong Wu[†]

This Version: September 21, 2026

Abstract: This paper develops a cognitive foundation for base-rate neglect within a unified framework spanning the standard description paradigm and an experience paradigm in which decision-makers learn the base rate from a sequence of observations. In the model, noisy coding compresses the represented base rate toward a neutral default, generating base-rate neglect. Coding precision depends on both individual numerical acuity and exposure to base-rate information. The model therefore predicts weaker neglect among decision-makers with sharper numerical acuity in both paradigms and, under experience, among those who collect larger samples. An online experiment supports the model's main predictions. Independently measured numerical acuity is associated with greater base-rate weighting under both Description and Experience. Under Experience, subjects draw fewer cards when the base rate is more extreme, and subjects who sample more intensively place greater weight on the base rate. Base-rate neglect remains substantial whether it is evaluated against the objective base rate or the frequency personally experienced by the decision-maker. Average base-rate weighting is estimated to be lower under Experience, but the difference is imprecisely estimated; Experience instead reveals substantially greater heterogeneity in base-rate weighting, associated with differences in sampling intensity.

Keywords: base-rate neglect; noisy coding; approximate number system; decisions from experience; belief updating

JEL Classification: C91, D83, D91

*ACKNOWLEDGMENT.

[†]Jilong Wu: $\text{Risla}\alpha\beta$, Department of Economics, Gent University, jilongwu@ugent.be.

1 Introduction

Base-rate neglect (BRN, hereafter) is one of the most extensively documented departures from Bayesian reasoning: in updating on new evidence, people give too little weight to prior information — the base rate — relative to Bayes’ rule. Early probability-inference experiments documented this underweighting of prior information (Phillips and Edwards, 1966), and Kahneman and Tversky (1973) brought the bias to prominence. It has since proved strikingly robust across tasks and incentivized replications (Grether, 1980; Koehler, 1996; Benjamin, 2019).¹ Yet, despite more than half a century of documentation, the cognitive mechanisms underlying BRN remain unsettled.

Further, almost all of this evidence comes from a single experimental *Description* paradigm. A decision-maker (DM) is told the base rate — as a probability (“15%”) or a natural frequency (“15 in 100”) — and combines it with a described signal. This Description paradigm has generated a large literature, and it remains the natural setting for the many applications in which base rates are stated explicitly. In a great many real-world settings, however, no one announces the base rate: people must learn it from experience. For example, an investor gauges how often a warning sign precedes a loss by living through markets. This is not a trivial difference: in risky choice, replacing described probabilities with sampled ones can reverse behavior outright (Baron and Erev, 2003; Hertwig et al., 2004; Wulff, Mergenthaler-Canseco and Hertwig, 2018). Whether the extent and heterogeneity of BRN change when the base rate must be learned by sampling — in the *experience paradigm* — rather than described, and if so why, has, to my knowledge, never been examined.

This paper addresses both gaps. I develop a noisy coding model of BRN that spans the two paradigms and test its predictions in an online experiment. The model builds on recent cognitive-psychology and economics work (Sanborn and Chater, 2016; Khaw, Li and Woodford, 2021; Enke and Graeber, 2023; Frydman and Jin, 2025; Oprea and Vieider, 2025). Under Description, a stated base rate is encoded with finite precision. Under Experience, the DM first acquires observations by sampling and then

¹Its most consequential form is diagnostic: told that a disease has a prevalence of 15% and that a test is 80% accurate, most people — physicians included — put the chance of illness after a positive result near the test’s accuracy, far above the correct value of 41% (Casscells, Schoenberger and Graboys, 1978; Esponda, Vespa and Yuksel, 2024; Eddy, 1982).

encodes the resulting frequency with finite precision. In both cases, imprecision leads her to combine the encoded information with a cognitive default, compressing the represented base rate toward that default. The framework therefore identifies a common mechanism for BRN while allowing information acquisition to play a distinct role under Experience.

A unified framework. The model starts from the noisy coding premise that a DM cannot represent numerical information with perfect precision (Khaw, Li and Woodford, 2021; Enke and Graeber, 2023; Frydman and Jin, 2025; Oprea and Vieider, 2025). Under *Description*, this constraint applies directly to the stated base rate. I formalize it using the Bayesian-sampler view of cognition (Sanborn and Chater, 2016; Zhu, Sanborn and Chater, 2020): the stated base rate is rendered into a finite internal sample. Under *Experience*, information acquisition is a separate first stage. The DM chooses how many external observations to draw and, after sampling ends, encodes the encountered frequency through the same finite-precision process. Thus the paradigms share a representational constraint, but Experience additionally makes the evidence that enters this constraint endogenous. In either case, the DM combines the noisy encoding with a cognitive default, generating compression toward that default.

Under *Description*, the precision of this internal representation is a limited cognitive resource — its concentration — determined by numerical acuity and exposure to the stated base rate. The DM combines the resulting finite internal sample with her cognitive default before incorporating the signal. BRN follows because the represented base rate is therefore compressed toward the default. A direct prediction is that subjects with sharper numerical acuity attach greater weight to the base rate, which I test using an independent dot-comparison measure of numerical acuity.

Under *Experience*, the DM must first draw observations herself. Under voluntary sampling, the amount of information the DM acquires is determined jointly by the subjective opportunity cost of further draws and the location of the base rate. The frequency she has encountered is then encoded before it enters posterior reasoning. Consequently, sampling behavior affects belief formation through exposure: conditional on numerical acuity, subjects who sample more intensively should place greater weight on the base rate.

Experimental evidence. I test these predictions in an online experiment with 198 subjects randomly assigned to Description or Experience. Subjects update beliefs about a binary state after seeing an informative signal; the base rate is either stated directly or learned by voluntary sampling. Under Description, subjects substantially underweight both the base rate and the signal: the estimated weights are 0.73 and 0.84, respectively. I measure numerical encoding precision with an independent nonsymbolic dot-comparison task. Subjects with sharper numerical acuity place significantly greater weight on the base rate, an effect robust to controlling for CRT and response time, while numerical acuity does not affect signal weighting.

Under Experience, subjects again underweight both sources of information: the estimated weights on the objective base rate and signal are 0.64 and 0.70, respectively. I also evaluate base-rate weighting using the experienced frequency in the subject's own sample. The estimated base-rate weight remains well below one under this alternative measure (0.68). Moreover, the numerical-acuity gradient reappears under Experience, so the central conclusion is robust to both notions of the base rate. Subjects also sample less at more extreme base rates, as the stopping model predicts. Across subjects, those who draw more cards place greater weight on the base rate, consistent with the model's exposure mechanism.

Comparing conditions reveals no precisely estimated average difference in base-rate weighting. The more pronounced contrast concerns heterogeneity. The distributions of individual base-rate weights differ across conditions: Experience has both a heavier lower tail of subjects who place very little weight on the base rate and a heavier upper tail of subjects who overweight it. Subjects in the Experience upper tail draw more cards on average, consistent with heterogeneity in sampling intensity contributing to this broader distribution. Signal weights are also lower under Experience, both on average and throughout most of their distribution.

Implications. These findings have three implications. First, the framework supplies a cognitive micro-foundation for BRN: rather than treating it as a fixed departure from Bayes' rule, the model derives its severity from the precision with which the base rate is encoded. This precision reflects a basic capacity for numerical encoding but is also shaped by environmental features that determine exposure to base-rate information.

The framework therefore treats BRN as endogenous and context-dependent, rather than as a fixed individual characteristic. Second, Experience should not be treated simply as another information format. It adds an information-acquisition stage before the later representation stage, a distinction that may also be relevant when base rates are ambiguous and people do not have exact information. A common constraint on numerical encoding provides one route to understanding heterogeneity across such information environments. Third, the model suggests a potential remedy for BRN. For example, certain training on the approximate number sense or structured exposure to a larger, representative sample may strengthen the representation of the base rate and thereby reduce neglect. Finally, although it is beyond the model’s primary focus, the weaker signal inference under Experience suggests that how people acquire and perceive base-rate information may also affect their use of other information. This interaction between cognitive components is an important direction for further work.

Related literature. Base-rate neglect has been studied in a large literature spanning many forms of the Bayesian updating task (see [Kahneman and Tversky, 1973](#); [Bar-Hillel, 1980](#); [Barbey and Sloman, 2007](#), for a few influential examples). Its severity ranges widely: some studies find the base rate discarded almost entirely ([Bar-Hillel, 1980](#)), while in the more typical case respondents do lean on it, only far too little relative to Bayes’ rule ([Koehler, 1996](#); [Benjamin, 2019](#)). Further, among the literature, the most famous finding is that presenting the same statistics as counts drawn from a population (“10 in 1000”) rather than as percentages (“1%”) roughly triples the rate of correct inference ([Gigerenzer and Hoffrage, 1995](#)). The gain is real but bounded — most people still underweight the base rate even in frequency form — so that format *attenuates* the bias without removing it ([Barbey and Sloman, 2007](#); [McDowell and Jacobs, 2017](#)).

Why BRN occurs, and why it moves with environmental features like format, remains debated. [Kahneman and Tversky \(1973\)](#) attributed the neglect to a representativeness heuristic, which highlights how diagnosticity can crowd out base rates but does not itself explain the format-dependent neglect observed in the data. A second, ecological-rationality tradition disputes that the phenomenon is a failure of reasoning at all: because the mind is adapted to the event *counts* of natural sampling rather

than to normalized probabilities, presenting the statistics as natural frequencies largely recovers the Bayesian answer, so the “bias” looks less like irrationality than like an intelligence tuned to its environment (Gigerenzer and Hoffrage, 1995). Barbey and Slovic (2007) contest this reading, arguing that frequencies help not because the mind computes in counts but because they expose the nested-set structure of the problem, prompting a shift from associative to rule-based reasoning. A related tradition instead holds that people integrate cues additively rather than multiplicatively, assigning the base rate too little linear weight (Juslin et al., 2011).

This paper contributes a formal model of BRN, grounded in the cognitive premise — shared by a growing body of work — that people cannot encode magnitudes, probabilities and frequencies among them, with full precision (Sanborn and Chater, 2016; Khaw, Li and Woodford, 2021; Enke and Graeber, 2023; Bouchouicha et al., 2025; Frydman and Jin, 2025; Oprea and Vieider, 2025). It is also related to accounts that trace bias to *random noise* — notably Costello and Watts (2014), in whose account reasoning obeys probability theory but is perturbed by noise that produces systematic bias. My model makes explicit both *what* is mis-encoded — the base rate — and *how* the error enters inference. Neglect is therefore not generic output noise but the result of an evidence weight determined by base-rate encoding precision. Chan et al. (2025) provide related evidence that greater exposure to an objectively given frequency grid increases the weight subjects place on the prior; their analysis emphasizes confidence over possible priors, whereas the present framework models the precision of the encoded base rate itself. Augenblick, Lazarus and Thaler (2025) provide a complementary account in which people imprecisely estimate signal strength, producing overinference from weak signals and underinference from strong ones. Ba, Bohren and Imas (2024) likewise model underreaction through imprecise belief representation in a multistate updating problem. This paper instead isolates the base-rate channel and extends it from stated to sampled information.

This study also belongs to the literature on the decision-from-experience paradigm, which has focused overwhelmingly on individual risk-taking. There, the well established *description–experience gap* shows that DMs can behave systematically differently when a lottery is learned by sampling rather than read from a summary: rare events that loom large when their probabilities are stated tend to be underweighted when

those same probabilities are sampled, so that the direction of probability-dependence reverses across the two paradigms (Barron and Erev, 2003; Hertwig et al., 2004; Wulff, Mergenthaler-Canseco and Hertwig, 2018). The gap is not solely a mechanical consequence of sampling error, and recent work attributes an important role to the noisy representation of sampled information (Oprea and Vieider, 2025). This program, however, has centered almost entirely on risk and lottery valuation. Belief updating, and BRN in particular, has by contrast been studied almost exclusively under Description. Whether BRN arises when the base rate itself is sampled, and how it is shaped by sampling behavior, remains unexamined.

The nearest evidence addresses a related but distinct question: whether accumulated experience mitigates BRN. For example, Araujo, Chabris and Meyer (2025) use the fact that medical professionals face diagnostic base-rate problems throughout their careers, and find that more experienced clinicians neglect base rates less — they are less likely to ignore prevalence outright, report more accurate posteriors, and track changes in the base rate more closely — with no comparable pattern among non-medical workers. Esponda, Vespa and Yuksel (2024) further reveal a pronounced pattern in the laboratory: across 200 rounds with outcome feedback, BRN recedes only slowly and partially, because confidence in one’s initial response limits how much later feedback is allowed to correct it. Both show that experience can reduce BRN, but unevenly: how much erodes depends on the confidence an initial understanding creates and how attentively feedback is processed, so the same amount of experience can leave very different residual neglect across contexts.

Their notion of experience, though, is long-run learning from repeated outcome *feedback* — years on the job, or hundreds of trials — not the single, feedback-free sample that defines the experience paradigm here. Many of the highest-stakes base-rate judgments, such as a juror’s verdict in a single trial or a one-time investment call, are made once, with no chance for that kind of feedback to operate; what matters there is whether sampling the base rate, rather than being told it, changes how it is used in the first place. I bring this setting, and its comparison with Description, to BRN. The results show that the paradigms differ not in a precisely estimated average degree of BRN but in its heterogeneity: individual base-rate weights are more heterogeneous under Experience.

2 Theoretical Framework

I proceed in two steps. First, in a basic setup where the base rate is given as explicit descriptive statistics (e.g., ten Orange cards and ten Blue cards), I show how the decision-maker (DM) encodes it under representational noise and how the resulting imprecise mental representation gives rise to base-rate neglect (BRN). The key object is a single encoding-precision parameter, which I link to the precision of the DM’s approximate number system. Second, I model the case in which the DM must instead learn the base rate through sampling. There the acquisition of evidence and its representation separate into two stages, generating predictions about her sampling behavior and about how both numerical acuity and sampling exposure map into the weight she places on the base rate.

2.1 Basic (Descriptive) Setup

I represent the DM’s belief about a binary state (State A versus State B) directly as a probability $p \in (0, 1)$, the probability that the true state is A . I restrict attention to this two-state case, on which the literature has focused. Nothing in what follows is special to two states: the same construction extends to $N > 2$ mutually exclusive states by replacing the Beta prior and Binomial encoding below with their Dirichlet and Multinomial counterparts (e.g., Ba, Bohren and Imas, 2024).

Mental default. Before processing the stated base rate, the DM holds a weak cognitive default over it,

$$p \sim \text{Beta}(a_0, b_0), \tag{1}$$

with $a_0 = b_0$ and a small $\kappa_0 = a_0 + b_0$. The equality centers the default at $p_0 = a_0/(a_0 + b_0) = 1/2$, in line with earlier work (Enke and Graeber, 2023) and with empirical evidence from a related belief-updating task (Ba, Bohren and Imas, 2024).² The concentration κ_0 determines how tightly the default is held around its mean. I treat the

²Related Bayesian-sampling accounts often impose the stronger normalization $\text{Beta}(1, 1)$, which sets $\kappa_0 = 2$ and corresponds to the uniform “ignorance prior” (e.g., Ba, Bohren and Imas, 2024). The theory does not need that exact value: any fixed, finite, low-concentration default produces the same compression logic. Nor is exact neutrality essential for underweighting; allowing $p_0 \neq 1/2$ mainly adds a default lean, which shows up as a nonzero intercept in the Grether mapping below.

default as fixed and focus on how encoding imprecision generates BRN.

Encoding. I assume that even an exactly stated probability is represented with finite precision. Numerical magnitudes — whether presented as symbols or as sets of items — are carried by noisy internal signals, a premise that underlies noisy-coding models of judgment and choice (Dehaene, 2003; Piazza et al., 2004; Khaw, Li and Woodford, 2021; Frydman and Jin, 2022; Enke and Graeber, 2023; Vieider, 2024; Frydman and Jin, 2025; Oprea and Vieider, 2025). I give this premise a form suited to a probability: the DM represents the stated base rate by a finite number κ_D of internal samples (D for description),

$$\hat{a} \sim \text{Binomial}(\kappa_D, p), \quad \hat{b} = \kappa_D - \hat{a}, \quad (2)$$

so that the encoded frequency $\hat{a}/\kappa_D$ is an unbiased but noisy signal of p : $\mathbb{E}[\hat{a}/\kappa_D \mid p] = p$.³ The parameter κ_D governs how noisy this signal is: the more internal samples, the more tightly the encoded frequency clusters around p (its variance is $p(1-p)/\kappa_D$). I therefore refer to κ_D as the DM’s encoding precision — the counterpart, on the side of the encoded signal, of the default’s concentration κ_0 (cf. Ba, Bohren and Imas, 2024).

The encoding precision depends on two factors. The first is the precision of the DM’s approximate number sense (ANS) — the cognitive system that represents numerical magnitude — which I index by θ . Numerosities recruit an approximate magnitude representation whether they are presented as symbols or as sets of items (Piazza et al., 2007), and individual differences in its precision are associated with symbolic numerical performance (Halberda, Mazzocco and Feigenson, 2008; Park and Brannon, 2013). The second is the exposure e the base rate receives, which depends both on the DM (e.g., how long and how attentively she processes it) and on the display (e.g., how readable the stimulus is). I therefore write

$$\kappa_D = \kappa(e; \theta), \quad (3)$$

and assume that κ is increasing in both arguments.

³This construction is consistent with the “Bayesian sampler” view that the brain approximates beliefs through finite internal samples (Sanborn and Chater, 2016; Zhu, Sanborn and Chater, 2020). One possible neural implementation treats the samples as the binary outputs of a finite population of neurons, in which case κ_D plays the role of neural capacity (Frydman and Jin, 2025).

Decoding. The internal counts $(\hat{a}, \hat{b})$ are a noisy, finite-sample signal of the base rate. The DM therefore combines them with the cognitive default. Under the assumed prior this operation minimizes mean squared error across repeated estimates: the reduction in variance more than offsets the bias introduced by regression toward the default (Khaw, Li and Woodford, 2021; Frydman and Jin, 2022; Vieider, 2024; Oprea and Vieider, 2025). Because Beta is conjugate to Binomial, the combination is exact Bayesian updating:

$$p \mid \hat{a} \sim \text{Beta}(a_0 + \hat{a}, b_0 + \hat{b}). \quad (4)$$

For a given set of internal samples, the decoded base rate is the posterior mean, $(a_0 + \hat{a})/(a_0 + b_0 + \kappa_D) = \gamma \hat{a}/\kappa_D + (1 - \gamma) p_0$. The realized representation varies with $\hat{a}$. Because the empirical predictions concern average reports, I focus on its expectation over the encoding noise and refer to this average as the represented base rate:

$$p_{BR} \equiv \mathbb{E} \left[\frac{a_0 + \hat{a}}{a_0 + b_0 + \kappa_D} \mid p \right] = \gamma p + (1 - \gamma) p_0, \quad \gamma = \frac{\kappa_D}{\kappa_D + \kappa_0} = \frac{1}{1 + \kappa_0/\kappa_D}. \quad (5)$$

The evidence weight γ weighs the κ_D internal samples against the κ_0 pseudo-observations of the default, so only their ratio matters. As encoding becomes precise, γ approaches 1 and the compression vanishes; as the internal signal becomes noisy, the default receives more weight.

Posterior beliefs. With the decoded base rate p_{BR} , the DM updates on a signal S of known likelihood ratio $\frac{p(S|A)}{p(S|B)}$. Bayes' rule states that posterior odds equal prior odds times this likelihood ratio:

$$\frac{p_{Post}}{1 - p_{Post}} = \frac{p_{BR}}{1 - p_{BR}} \cdot \frac{p(S \mid A)}{p(S \mid B)}. \quad (6)$$

Taking logs gives the log-odds-additive form:

$$\ell_{Post} = \ell_{BR} + \ell_S, \quad \ell_{BR} = \ln \frac{p_{BR}}{1 - p_{BR}}, \quad \ell_S = \ln \frac{p(S \mid A)}{p(S \mid B)}. \quad (7)$$

This expression maps naturally, though not exactly, onto the regression of Grether (1980), which relates elicited posterior log-odds to the signal and prior log-odds:

$$\ell_{Post} = \beta_0 + \beta_1 \ell_S + \beta_2 \ell, \quad \ell = \ln \frac{p}{1 - p}. \quad (8)$$

Viewed through the model, each coefficient has a model-based interpretation. The signal coefficient β_1 is the weight on the signal's log-likelihood ratio ℓ_S , which the model

leaves unrestricted, as signal processing is beyond its present scope.⁴ The intercept β_0 captures a standing lean toward one state: it is zero under the neutral default and nonzero otherwise (Appendix B).

The base-rate coefficient β_2 is increasing in the evidence weight γ . In probability space this relationship is exact: by (5), the represented base rate moves with the true one at the constant rate

$$\frac{\partial p_{BR}}{\partial p} = \gamma. \quad (9)$$

Because the Grether regression is estimated in log-odds, the fitted coefficient is not mechanically equal to γ : the logit transformation makes the map from ℓ to ℓ_{BR} curved. The two coincide locally at $p = 1/2$, and a larger γ yields a larger fitted β_2 in the symmetric design used here. Appendix B gives the exact mapping.

Prediction 1 (ANS precision and BRN). *The more precisely the DM encodes numerical magnitude, the larger her evidence weight γ and the less her represented base rate is compressed toward the neutral default. Thus DMs with a more precise ANS (higher θ) should exhibit weaker base-rate neglect.*

Relation to existing findings. Existing evidence speaks mainly to the exposure argument *e*. Chan et al. (2025) find that DMs place more weight on a prior after viewing its underlying frequency grid for longer, consistent with greater exposure raising κ_D and hence γ . The same channel may help explain why natural-frequency formats reduce BRN if they make the base rate easier to encode (Gigerenzer and Hoffrage, 1995). The other argument of $\kappa(e; \theta)$ — individual numerical acuity — has received less attention.

2.2 The Experience Paradigm

In the basic setup the base rate is handed to the DM, and all she must do is render a stated number into a finite set of internal samples. I now turn to the experience paradigm in which she gathers the evidence herself, drawing observations one at a time and deciding when to stop.

Voluntary sampling. Let $s = a + b$ be the number of draws so far, a of them favoring State A , and $\hat{p}_s = a/s$ the running frequency. The DM monitors how much this running

⁴Noisy signal representation could be modeled in parallel to the base-rate channel developed here; see Ba, Bohren and Imas (2024) for one such approach.

frequency can still move. Its sampling dispersion is

$$\widehat{V}_s = \frac{\hat{p}_s(1 - \hat{p}_s)}{s}, \quad (10)$$

and I use a one-step-lookahead rule under which another draw is worthwhile when its expected reduction in this dispersion exceeds its cost (Vieider, 2026). This is a myopic rule: it compares one additional draw with stopping immediately rather than solving a full dynamic program. To leading order, holding the current frequency locally fixed, the reduction in (10) from one more draw is $\hat{p}_s(1 - \hat{p}_s)/s^2$; Appendix C gives the exact one-step calculation. Writing $c > 0$ for the DM's subjective opportunity cost of another draw, she continues while $\hat{p}_s(1 - \hat{p}_s)/s^2 \geq c$. The resulting approximate target sample size is

$$s^* = \sqrt{\frac{\hat{p}(1 - \hat{p})}{c}}. \quad (11)$$

The sample size s^* is thus *endogenous to the observed frequency*. The DM stops when further drawing no longer pays, and the value of another draw depends on how much the running frequency can still move. This is greatest when the deck is balanced and least when it is lopsided. At an extreme base rate she is likely to encounter a near-uniform stream, so $\hat{p}(1 - \hat{p})$ falls quickly and sampling ends sooner; near one half, draws tend to remain balanced and sampling continues longer.

Prediction 2 (Endogenous stopping). *Under experience, expected sample size is endogenous to the base rate: on average, the DM draws fewer observations for more extreme base rates, producing an inverted-U in sample size that is symmetric about $p = 1/2$.*

Representation: encoding the drawn sample. Once sampling ends, the encountered frequency is a numerical input just as the stated composition was under description. Having stopped at s^* with realized frequency $\hat{p}$, the DM must render that input into a reportable magnitude. The same representational constraint therefore applies, although the source is now a sequence of observations and the number of draws s measures the DM's exposure to it. She represents $\hat{p}$ by κ_E internal samples, whose number is the encoding function evaluated at s ,

$$\kappa_E = \kappa(s; \theta), \quad (12)$$

where the subscript E denotes experience, so that drawing more cards supports a sharper representation.

The encountered frequency is encoded as

$$\hat{a}_E \sim \text{Binomial}(\kappa_E, \hat{p}), \quad \hat{b}_E = \kappa_E - \hat{a}_E, \quad (13)$$

rather than carried into inference as the literal counts a and b . Decoding then proceeds as in (5): combining the internal samples with the cognitive default and averaging over the encoding noise yields the average represented base rate

$$p_{BR,E} = \gamma_E \hat{p} + (1 - \gamma_E) p_0, \quad \gamma_E = \frac{\kappa_E}{\kappa_E + \kappa_0}, \quad (14)$$

and the same log-odds-additive posterior (7) as before.

Base-rate neglect under experience is therefore governed by the same two forces as under description. A DM with a more precise ANS obtains a higher encoding precision at any level of exposure. Exposure, however, is now the outcome of the stopping rule (11): the DM supplies her own. Holding θ fixed, heterogeneity in the cost of drawing c generates a positive cross-sectional association between sample size and γ_E . Because sample size is chosen, this association does not identify the causal effect of an exogenously assigned sample size.

Prediction 3a (ANS precision under experience). *Under experience the base-rate weight γ_E is increasing in the precision of the DM's ANS, θ . Base-rate neglect is therefore weaker among DMs with a more precise ANS, as under description.*

Prediction 3b (Exposure under experience). *Under experience the evidence weight γ_E is increasing in the DM's exposure to the deck, measured by her sample size s . Conditional on encoding precision, base-rate neglect is therefore weaker among DMs who draw more.*

3 Experimental Design

Experiments on base-rate neglect typically induce a base rate, condition beliefs on a signal, and then elicit the posterior belief (Kahneman and Tversky, 1973; Grether, 1980). My design follows the same logic, but implements it in two paradigms. In the Description condition, the base rate is stated explicitly as a natural frequency. In the Experience condition, the same base rate is hidden and must be learned by drawing cards from the deck. The two conditions therefore share the same inference problem and differ in how participants learn the base rate.

3.1 Task Environment

Each trial implements a binary-state inference problem in which the participant uses a weather signal to infer which city was selected. The state is determined by a card drawn from a 20-card deck containing Orange City and Blue City cards. As shown in Figure 1, the deck contains k Orange City cards and $20 - k$ Blue City cards, so $k/20$ is the base rate. Participants are told that each round uses a new deck and that every deck contains at least one card of each city. Conditional on the selected city, the computer draws one day uniformly from that city’s 10-day weather record. Orange City’s record contains n sunny days and $10 - n$ rainy days, while Blue City’s record is the mirror image, with $10 - n$ sunny days and n rainy days ($n > 5$). The two records therefore define the signal structure, and the weather of the drawn day is the signal.

The elicitation is implemented by the strategy method (Brandts and Charness, 2011; Chan et al., 2025): before learning which day was actually drawn, the participant reports the posterior probability assigned to each city once conditional on a sunny day and once conditional on a rainy day. The order of these two contingent reports is randomized within each trial. Each report uses two linked sliders, one for each city, with no default position. The report range is restricted to 1–99, so the participant cannot report 0% or 100%; this is consistent with the instruction that the deck always contains at least one card from each city. Moving either slider automatically moves the other in the opposite direction, so the two reported probabilities always sum to 100%.

Stimuli. Across trials I vary the deck’s composition k , spanning extreme base rates to more moderate ones ($k \in \{1, 2, 4, 6, 8, 10, 12, 14, 16, 18, 19\}$, i.e. base rates from 0.05 to 0.95). Varying the base rate within participant identifies its weight in the posterior, the coefficient β_2 in the Grether regression (8). The weather record also varies across trials ($n \in \{6, 7, 8, 9\}$): a sunny day favors Orange City by $n : (10 - n)$, while a rainy day favors Blue City by the same ratio. This signal variation identifies the likelihood weight β_1 . Table A.2 lists the full set of stimuli: eleven base-rate levels at the fixed signal structure $n = 7$, plus three further signal structures ($n = 6, 8, 9$) at the fixed moderate base rate $k = 10$ (a base rate of 0.5).

The deck this round:

6 **Orange City** cards and 14 **Blue City** cards.

Here are the two weather records:

The Orange City

7 sunny days; 3 rainy days

The Blue City

3 sunny days; 7 rainy days

The computer has already drawn a card, and then a day from the record of the city on that card.

1 of 2

Suppose the day the computer drew turns out to be

sunny ☀

In that case, how likely is it that the card the computer drew was an **Orange City** card or a **Blue City** card?

68% **Orange City**

32% **Blue City**

Figure 1: Screenshot of the posterior-belief sliders. *Note:* The participant reports the posterior probability assigned to each city using two linked sliders. Moving one slider automatically updates the other, so the two probabilities always sum to 100%.

3.2 Description Condition

In the Description condition, participants are told the deck’s composition and each city’s weather record, but not the card selected by the computer or the day that will determine payment. Both pieces of environment-level information are conveyed as natural frequencies, making the Description condition more directly comparable to the Experience condition. The deck composition is stated as “The deck this round: k Orange City cards and $20 - k$ Blue City cards,” and each city’s record is stated in the same count format, for example “Orange City: n sunny days; $10 - n$ rainy days.” Figure 1 shows an example task screen.

3.3 Experience Condition

The Experience condition keeps the same environment — the two cities, their weather records, the two contingent signal questions, and the elicitation sliders — with one exception: participants are no longer told the deck’s composition. Instead, as shown in Figure A.1, they learn it by drawing cards from the 20-card deck themselves, one at a time, with replacement, and stop whenever they judge that they have learned enough by clicking Stop. They may draw as many or as few cards as they like, subject only to a floor of three draws.⁵ The base-rate and signal-structure levels crossed within participant are otherwise the same as in the Description condition.

3.4 Incentives

In addition to the fixed participation fee of \$5, participants receive a bonus. The bonus starts at \$2.50 and falls by \$0.25 for every percentage point this report lies from the statistically correct posterior, floored at \$0. Because only one randomly selected trial and one contingent report within that trial are ever paid, no report can be hedged against another, and the best strategy is to report the honest posterior for both signal realizations in every trial, as stressed in the experimental instruction. I favor this simple, fully transparent rule over more complex incentive-compatible mechanisms such as BDM because recent evidence shows the latter can themselves bias truthful reporting (Danz, Vesterlund and Wilson, 2022); see also recent similar practice in Enke, Graeber and Oprea (2023) and Ba, Bohren and Imas (2024).

3.5 Measuring Encoding Precision

To measure how precisely a participant encodes numerical magnitude, participants complete a nonsymbolic dot-comparison task after the main task (Halberda, Mazzocco and Feigenson, 2008). On each trial, two intermixed sets of colored dots flash briefly on

⁵Each drawn card is shown on its own, large and centered, and the draw button freezes briefly after every draw, so that the participant attends to each card individually rather than to a running tally. No cumulative count of Orange City versus Blue City cards is shown, and the drawn cards are not displayed again once the participant stops, so the report rests on what the participant carries in memory. Drawing has no bearing on the computer’s hidden draw or on the day’s weather, which are generated exactly as in the Description condition.

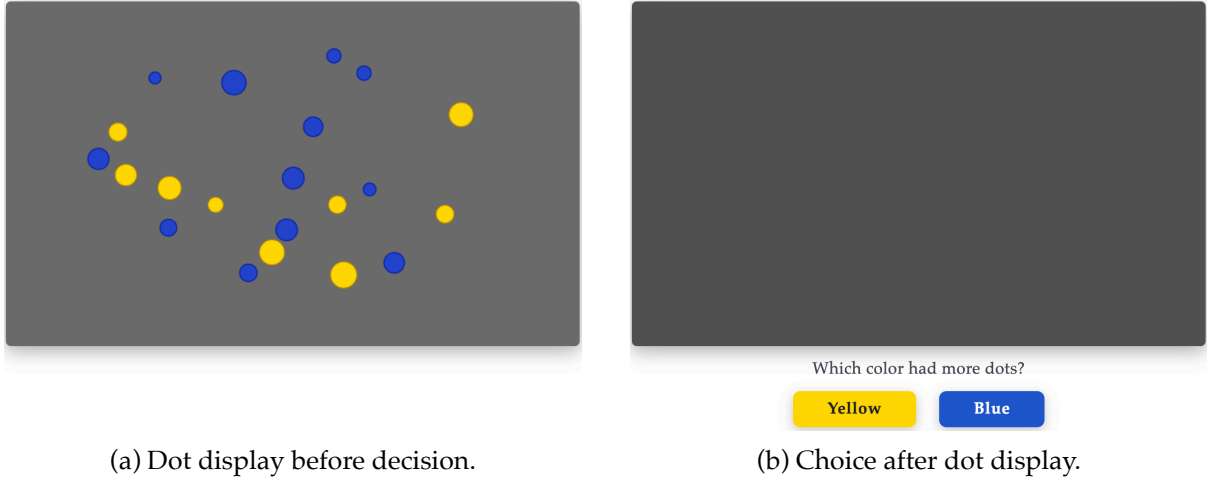


Figure 2: Screenshot of an example dot-comparison trial. *Note:* Panel (a) shows the dot display, in which two intermixed sets of colored dots appear briefly on screen; panel (b) shows the response screen, where the participant reports which color was more numerous.

screen and the participant judges which color is more numerous (Figure 2).⁶ Because the ratio of the two counts varies across trials, the task reveals how finely a participant discriminates magnitudes — a stable individual characteristic of her ANS (Halberda, Mazzocco and Feigenson, 2008). I summarize it by her Weber fraction $\hat{w}$, estimated from the standard psychophysical model of her choices (see details in Appendix D). Unlike the raw share of correct answers, which pools near-certain easy trials with near-chance hard ones, $\hat{w}$ uses the whole psychometric curve and is comparable to prior estimates. I refer to the standardized negative of $\hat{w}$ as the participant’s *numerical acuity*, so that higher values indicate sharper acuity.

I use this acuity as an empirical proxy for the ANS precision θ in the model. The maintained hypothesis is that the dot task and the belief task draw on a common numerical representation, so a participant who discriminates numerosities more finely also encodes base rates less noisily.⁷ Because it is measured outside the belief task — in a fraction-of-a-second perceptual judgment rather than a deliberative probability report — the measure is less directly exposed to the belief task’s priors and response habits. A correlation between the two would then be consistent with the common encoding-precision channel the model posits, though not by itself decisive without considering natural confounds, such as cognitive ability, attention, and effort.

⁶The array sizes, count ratios, display timing, and the acuity-estimation procedure are relegated to Appendix D.

⁷See Khaw, Li and Woodford (2021) for a detailed discussion of how the noisy coding of numerical magnitude can serve as a common mechanism across perceptual and economic judgments.

3.6 Implementation Details

I ran the experiment online on Prolific, restricting eligibility to native English speakers with a high prior approval rate ($> 99\%$). Participants were randomly assigned to either the Description or Experience condition. Each participant first read the instructions, which included simple worked examples, then completed one unincentivized practice round to become familiar with the interface, and finally took a four-question comprehension test covering the deck composition, the two cities' weather records, the meaning of the signal, and the payment rule.⁸ Participants who failed the comprehension check exited the study and received \$0.50 for partial participation.

In the main belief task each participant completed 15 task rounds: 14 distinct rounds presented in randomized order, followed by one repeated round at the end to assess the overall reliability of elicitation. Each round varied the deck composition and/or the weather records and elicited both contingent signal reports one after the other, with the order of the sunny and rainy questions randomized across rounds. Participants received no feedback during the task; at the end, they learned the realized card and weather only for the randomly selected round used to determine payment. After the belief task, participants completed the dot-comparison task: a block of 40 trials preceded by brief practice, with a second 40-trial block administered in the Description condition to make total task duration comparable across conditions, since drawing cards makes the Experience condition longer. All numerical-acuity analyses use responses from the first block in both conditions; under Description, the responses in the two blocks are statistically indistinguishable. At the end, participants answered a short questionnaire on basic characteristics.

4 Results

This section presents the experimental results. Subsection 4.1 documents base-rate neglect (BRN) under Description and tests whether it varies with numerical acuity. Subsection 4.2 turns to the Experience condition and examines how subjects sample and

⁸In the Experience condition, the comprehension test replaces the question on the deck's composition with a question on how the participant finds out the deck's composition, reviewed against the sampling screen rather than the report screen; the remaining three questions are unchanged.

how numerical acuity and sample size shape their neglect. Subsection 4.3 compares the two conditions.

Sample. I collected 198 subjects’ complete responses: 97 (101) subjects were randomly assigned to the Description (Experience) condition. Median completion time was 25.2 minutes in Description and 26.8 minutes in Experience. Appendix Table A.1 reports subject characteristics across the two conditions, which are similar overall.

4.1 BRN under Description

Overall pattern. Figure 3a plots reported posterior log-odds, ℓ_{Post} , against prior log-odds, ℓ , in the base-rate variation block, separately by signal. In these coordinates Bayesian updating is a line with slope one, so BRN appears as a reported line flatter than the benchmark. The data show this regressive pattern: reports lie above the Bayesian posterior at low base rates and below it at high base rates.⁹ Column (1) of Table 1 quantifies this with the Grether regression (8), estimated on all rounds: the base-rate coefficient is $\beta_2 = 0.728$, well below the Bayesian benchmark of one; the signal coefficient is also below one ($\beta_1 = 0.840$), so subjects underweight the signal as well. Formal tests reject both $\beta_1 = 1$ and $\beta_2 = 1$ ($p < 0.01$). The constant β_0 is small and statistically insignificant, consistent with the neutral default ($a_0 = b_0$) in (1).

Result 0. *Under Description, subjects underweight the base rate, as widely documented in the literature.*

Numerical acuity. A dot-comparison task gauges how precisely each subject represents numerosity. Figure 3b plots the share of trials on which yellow is chosen as the more numerous color against the log-ratio $\log(n_{\text{yellow}}/n_{\text{blue}})$. Choices trace the S-shaped curve typical of this task: accuracy is modest when the two sets are close in number and near ceiling when one is twice the other, rising from 67% at 8:7 to 94% at 2:1. The standard model of this task (Halberda, Mazzocco and Feigenson, 2008) fits these choices with a group Weber fraction of $w = 0.27$.¹⁰ To check that the task measures numeri-

⁹The sunny- and rainy-signal lines run parallel (see Figure 3a), so the degree of BRN does not depend on which signal is realized. Regressing ℓ_{Post} on ℓ , a sunny-signal dummy and their interaction in this block, with standard errors clustered by subject, the interaction is essentially zero (-0.003 , SE 0.025 , $p = 0.91$).

¹⁰This acuity is broadly in line with prior work using this task (Halberda, Mazzocco and Feigenson, 2008; Halberda and Feigenson, 2008), though coarser than the average reported for laboratory adults,

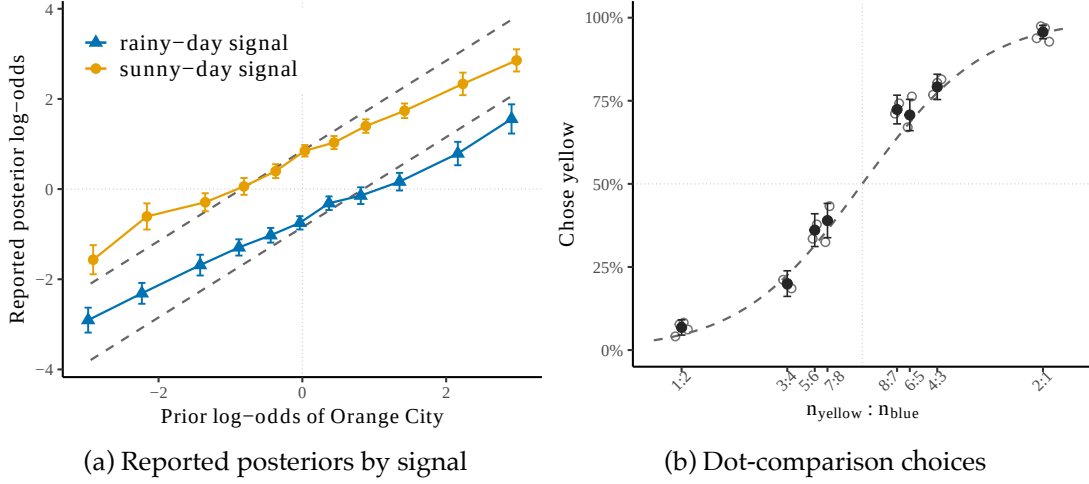


Figure 3: Belief reports and numerical acuity under Description. *Note:* Description condition, $N = 97$. Panel (a): mean reported posterior log-odds against prior log-odds in the base-rate variation block (likelihood 0.7), by signal; dashed lines are the Bayesian posteriors. See corresponding statistics in Table A.2. Panel (b): share of trials choosing yellow against the ratio $n_{\text{yellow}} : n_{\text{blue}}$ (log scale); filled points average across subjects, open circles split each ratio by the absolute count difference, and the dashed curve is the fitted ANS model (Weber fraction $w = 0.27$). 95% confidence intervals across subjects.

Table 1: Grether regression in the Description condition.

	(1)	(2)	(3)
β_0 (constant)	0.028 (0.017)	0.028 (0.017)	0.028 (0.017)
β_1 (signal, ℓ_S)	0.840*** (0.044)	0.840*** (0.044)	0.840*** (0.044)
β_2 (base rate, ℓ)	0.728*** (0.033)	0.728*** (0.032)	0.729*** (0.031)
$\ell \times$ numerical acuity		0.104*** (0.024)	0.096*** (0.024)
$\ell \times$ CRT			0.013 (0.034)
$\ell \times$ page time (between)			0.069** (0.032)
$\ell \times$ page time (within)			-0.010 (0.010)
p -value, $H_0: \beta_1 = 1$	< 0.01	< 0.01	< 0.01
p -value, $H_0: \beta_2 = 1$	< 0.01	< 0.01	< 0.01
Subjects	97	97	97
Observations	2716	2716	2716

Note: Column (1) is the baseline Grether regression (8); column (2) interacts numerical acuity (the standardized negative Weber fraction) with the prior; column (3) adds interactions of the prior with CRT and with between- and within-subject page time in seconds. These columns report interactions only, with moderator main effects omitted; the full specification including all main effects is Table A.4. Standard errors, clustered by subject, in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

cal discrimination rather than subject characteristics, I fit trial-level logits of a correct choice on the log ratio of the larger to the smaller set and on subject characteristics (see Appendix Table A.3). A larger ratio sharply raises the odds of a correct choice. Holding the ratio fixed, neither CRT nor gender, age, or education predicts accuracy in either condition, and none shifts sensitivity to the ratio. Thus, within these observed characteristics, numerical acuity is not systematically related to cognitive reflection or demographics.

Numerical acuity predicts BRN. Prediction 1 holds that, all else equal, more precise numerical encoding produces milder BRN. I test it by interacting the prior log-odds in the Grether regression with a subject’s numerical acuity — the standardized negative Weber fraction from the dot-comparison task, so that higher values mean sharper acuity. Column (2) of Table 1 bears out the prediction: the interaction is 0.104 ($p < 0.01$), so a one-standard-deviation increase in acuity raises the base-rate weight by about 0.10. Acuity does not, by contrast, change the weight on the signal (interaction -0.002 ; Table A.4), so sharper encoding raises the weight on the base rate specifically rather than on evidence in general. A natural concern is that the acuity gradient reflects something other than encoding precision, such as general cognitive ability or the effort a subject puts into the experiment. Column (3) therefore adds interactions of the prior with CRT and with page time, split into a subject’s average time per round and her round-to-round deviation from it. CRT has no bearing on BRN. Subjects who spend longer per round on average show significantly weaker neglect (0.069, $p < 0.05$), whereas a subject’s own slower rounds do not. This pattern is consistent with the exposure channel in (3): more time spent processing the base rate raises encoding precision. Most importantly, these controls leave the acuity interaction almost unchanged (0.096, $p < 0.01$).

Result 1. *As the model predicts, subjects with sharper numerical acuity exhibit weaker base-rate neglect.*

4.2 Sampling and BRN under Experience

Sampling behavior. Under Experience, subjects learn the base rate by sampling, so the first question is how much they choose to draw. Across subjects the mean sample size over the eleven base-rate rounds is 12.3 cards, with a median of ten. The sampling pat-

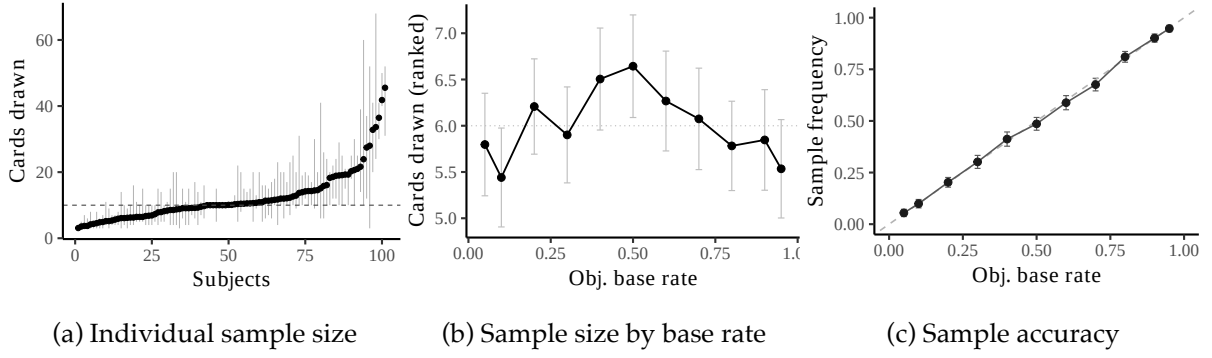


Figure 4: Individual sampling behavior. *Note:* Experience condition, base-rate-variation block, retest round excluded. Panel (a) shows how much subjects sample overall: each point is a subject’s mean number of cards drawn (subjects ordered along the axis by that mean), and the grey bars give her round-by-round min-max range; the dashed line is the grand median ($n = 10$). Panel (b) shows how sampling varies with the base rate within subject: the number of cards drawn is ranked within each subject’s eleven rounds and averaged across subjects at each base rate; the dotted line at the middle rank of 6, and the bars are 95% confidence intervals (± 1.96 standard errors). Panel (c) plots the sampled frequency, $\hat{p} = n_{\text{orange}}/n_{\text{draws}}$, against the true base rate, with ± 1.96 standard-error bars and the 45° identity dashed.

tern reveals substantial heterogeneity between and within subjects (Figure 4a): some subjects only draw a few cards close to the three-card floor in each round, while about half (51%) average more than ten cards per round and 12% average more than twenty. Some subjects (about 8%) appear to follow a fixed rule, drawing close to ten cards in every round, but most vary their sample size widely from round to round. Sampling is, moreover, unrelated to numerical acuity, consistent with the model. The rank correlation between a subject’s numerical acuity and her average number of draws is small and insignificant (Spearman $\rho = 0.15$, $p = 0.14$), and acuity does not predict how sharply sampling responds to the base rate.

The next question is whether subjects sample more when the base rate is closer to 0.5, as Prediction 2 predicts. Subjects differ markedly in their overall sampling levels, so I use within-subject ranks to isolate how they allocate sampling across rounds. I rank each subject’s eleven sample sizes from smallest (1) to largest (11) and average these within-subject ranks across subjects at each base rate (Figure 4b); with no relationship, the expected rank is the midpoint, 6. This is also the within-subject transformation used for the draw-rank control in Table 2. The pattern matches the prediction. Regressing the within-subject rank on the distance of the base rate from 0.5, $|p - 0.5|$, with standard errors clustered by subject, gives a coefficient of -2.15 ($p < 0.01$): moving from $p = 0.5$ to the most extreme base rates (0.05 or 0.95) lowers a round’s rank by about one

position.¹¹ Consistently, at the four most extreme base rates (0.05, 0.1, 0.9 and 0.95) the mean rank is 5.65, below 6 ($p < 0.01$), whereas at the three central ones (0.4, 0.5 and 0.6) it is 6.47, above 6 ($p < 0.01$). Sampling is thus heaviest where the deck is least lopsided.

Result 2. *Sampling is endogenous to the base rate of the round: subjects draw more cards when the base rate is closer to 0.5 and fewer toward the extremes.*

Before turning to how subjects use their draws, it is worth checking whether the sampling process displays a systematic aggregate directional bias. Because cards are drawn with replacement, the relative frequency of orange cards in a round should on average equal the true base rate. Figure 4c shows that the mean sampled frequency lies close to the 45° identity at every base rate. Thus, although individual samples remain noisy, the realized samples display no systematic aggregate directional bias. Plotting the median instead of the mean (Figure A.2) yields a similar pattern except at the two most extreme base rates ($p = 0.05$ and $p = 0.95$), where discreteness of short samples matters most.

BRN under Experience. The primary base-rate measure is the round’s *objective probability* — the true deck composition that generated the draws and defines the Bayesian benchmark, exactly as under Description. Column (1) of Table 2 gives a base-rate weight of $\hat{\beta}_2 = 0.641$, well below one ($p < 0.01$); the signal weight is likewise below one ($\hat{\beta}_1 = 0.702$, $p < 0.01$). Thus, subjects underweight both sources of information relative to the Bayesian benchmark.¹² Figure A.3 and Appendix Table A.5 report the corresponding cell-level posteriors.

For comparison, I also examine each subject’s *experienced frequency* — the relative frequency of the two cities among the cards she drew — as a secondary measure of the information available to her, as is common in the risky-choice literature (e.g., [Abdel-laoui, L’Haridon and Paraschiv, 2011](#)).¹³ The estimated base-rate weight is $\hat{\beta}_2 = 0.679$

¹¹A regression of the log number of cards drawn on $|p - 0.5|$ gives the same conclusion: the coefficient is -0.154 ($p < 0.05$), so moving from $p = 0.5$ to 0.05 or 0.95 reduces the number of draws by about 7%.

¹²Restricting the objective-base-rate regression to the interior sample used for the experienced-frequency analysis lowers the base-rate coefficient, because it removes many extreme-base-rate observations, but leaves the qualitative conclusions below unchanged. See Table A.6 in Appendix.

¹³When a subject draws only one colour the sampled frequency is 0 or 1 and its log-odds is undefined, so the experienced-frequency regressions use the interior sub-sample of rounds that drew at least one card of each city ($0 < n_{\text{orange}} < n_{\text{draws}}$), about 82% of all rounds. The excluded rounds fall almost entirely at the most extreme base rates ($p = 0.05$ and $p = 0.95$).

Table 2: Grether regression in the Experience condition.

	Obj. base rate			Sampled frequency		
	(1)	(2)	(3)	(4)	(5)	(6)
β_0 (constant)	-0.019 (0.029)	-0.019 (0.029)	-0.019 (0.029)	0.000 (0.030)	0.000 (0.030)	-0.002 (0.030)
β_1 (signal, ℓ_S)	0.702*** (0.057)	0.702*** (0.057)	0.702*** (0.057)	0.687*** (0.060)	0.687*** (0.060)	0.687*** (0.060)
β_2 (base rate, ℓ)	0.641*** (0.046)	0.641*** (0.044)	0.638*** (0.041)	0.679*** (0.051)	0.672*** (0.050)	0.638*** (0.048)
$\ell \times$ numerical acuity		0.117*** (0.027)	0.093*** (0.026)		0.096*** (0.029)	0.071** (0.028)
$\ell \times$ CRT			-0.026 (0.046)			0.006 (0.057)
$\ell \times$ page time (between)			-0.048 (0.051)			-0.064 (0.053)
$\ell \times$ page time (within)			-0.012 (0.014)			0.006 (0.024)
$\ell \times$ draws (between)			0.187*** (0.048)			0.131** (0.051)
$\ell \times$ draws rank (within)			-0.027 (0.031)			-0.049 (0.031)
p -value, $H_0: \beta_1 = 1$	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01
p -value, $H_0: \beta_2 = 1$	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01
Subjects	101	101	101	101	101	101
Observations	2828	2828	2828	2322	2322	2322

Note: The true base-rate log-odds in columns (1)–(3) are estimated on the full sample; the experienced frequency in columns (4)–(6) is estimated on the interior sub-sample of rounds that drew at least one card of each city (so $f \in (0, 1)$ and $\hat{\ell}$ is finite). Columns (1) and (4) are the baseline Grether regression (8). Columns (2) and (5) add the interaction of the prior variable with numerical acuity (the standardized negative Weber fraction). Columns (3) and (6) further add its interactions with CRT, with between- and within-subject page time in seconds, and with the subject’s sample size, entered as her mean number of draws per round between subjects and, within subject, as the rank of the round’s draw count among her own rounds. All moderators are standardized. Standard errors, clustered by subject, in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

(column 4), again below one ($p < 0.01$). Because this specification uses a selected sub-sample and a noisy, endogenous regressor, it is not mechanically comparable to the primary estimate; it nevertheless yields the same qualitative conclusion of substantial base-rate neglect.

The same acuity gradient found under Description reappears under Experience, and for both prior measures. Interacting the base-rate variable with numerical acuity gives a positive coefficient throughout: 0.117–0.126 when the prior is the objective probability (Table 2, columns 2–3) and 0.09–0.10 when it is the experienced frequency

(columns 5–6), all significant at the 5% level at least. Subjects with sharper numerical acuity thus weight the base rate more heavily, and the pattern survives controlling for CRT, response time, and sample size. Its appearance under both measures is consistent with numerical acuity governing the use of base-rate information, rather than reflecting only how well subjects read a stated probability.

Result 3. *Under Experience, base-rate neglect is substantial under the objective base-rate benchmark and also under the secondary experienced-frequency measure; under either measure it weakens with a subject’s numerical acuity.*

In addition to numerical acuity, the model predicts that greater exposure is associated with more precise base-rate encoding. Columns (3) and (6) of Table 2 show this cross-subject association: a subject who draws one standard deviation more cards per round on average (about eight cards) places 0.187 more weight on the objective base rate ($p < 0.01$) and 0.131 more on experienced frequency ($p < 0.05$). Page time does not predict base-rate weighting once sample size is controlled for. Within subjects, however, rounds in which they draw more show no greater weight on the base rate. Since sample size is chosen and jointly determined with the realized sample, the cross-subject association does not identify the causal effect of exogenously increasing draws; it is nevertheless consistent with the model’s exposure channel. Acuity is also unrelated to sampling intensity, consistent with the two sources of heterogeneity entering $\kappa(s; \theta)$ separately.

Result 4. *Under Experience, subjects who draw more cards on average weight the base rate more heavily, an association distinct from numerical acuity but not a causal exposure effect.*

4.3 Description versus Experience

Lastly, this subsection examines whether the description-experience gap also surfaces in this belief-updating task. I pool the responses from both conditions and re-estimate the Grether regression, interacting an Experience indicator with the signal and base-rate log-odds so that both weights are free to differ across conditions; the interaction terms then measure the Description–Experience differences. In the primary objective-base-rate specification, Experience is associated with a lower base-rate weight of 0.087 (SE 0.056), but this estimate is imprecise and does not reject equal base-rate weighting

Table 3: Grether regression, Description and Experience pooled.

	Obj. base rate			Sampled frequency		
	(1)	(2)	(3)	(4)	(5)	(6)
β_0 (constant)	0.028 (0.017)	0.028 (0.017)	0.028 (0.017)	0.028 (0.017)	0.028 (0.017)	0.028 (0.017)
β_1 (signal, ℓ_S)	0.840*** (0.044)	0.840*** (0.044)	0.840*** (0.044)	0.840*** (0.044)	0.840*** (0.044)	0.840*** (0.044)
$\ell_S \times$ Experience	-0.138* (0.072)	-0.138* (0.072)	-0.138* (0.072)	-0.153** (0.075)	-0.153** (0.075)	-0.153** (0.075)
β_2 (base rate, ℓ)	0.728*** (0.033)	0.728*** (0.032)	0.729*** (0.031)	0.728*** (0.033)	0.728*** (0.032)	0.728*** (0.031)
$\ell \times$ Experience	-0.087 (0.056)	-0.087 (0.054)	-0.087 (0.053)	-0.049 (0.061)	-0.056 (0.059)	-0.066 (0.057)
Experience	-0.047 (0.034)	-0.047 (0.034)	-0.047 (0.034)	-0.029 (0.035)	-0.028 (0.035)	-0.030 (0.035)
$\ell \times$ numerical acuity		0.110*** (0.018)	0.112*** (0.018)		0.101*** (0.018)	0.098*** (0.019)
$\ell \times$ CRT			-0.004 (0.029)			0.013 (0.029)
$\ell \times$ page time (between)			0.069** (0.029)			0.054** (0.027)
$\ell \times$ page time (within)			-0.014 (0.009)			-0.006 (0.011)
Subjects	198	198	198	198	198	198
Observations	5544	5544	5544	5038	5038	5038

Note: Both conditions pooled. β_0 , β_1 and β_2 refer to Description; the rows interacted with *Experience* (= 1 under Experience) give the differences under Experience. Under Experience, the prior is the objective base rate in columns (1)–(3) and the subject’s sampled frequency $\hat{p} = n_{\text{orange}}/n_{\text{draws}}$ in columns (4)–(6), the latter restricted to rounds with at least one card of each city; under Description it is always the objective base rate. Columns (1) and (4) are the baseline Grether regression (8); columns (2) and (5) add the interaction of the prior with numerical acuity; columns (3) and (6) add its interactions with CRT and with between- and within-subject page time. Moderators are standardized within each condition. Standard errors, clustered by subject, in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

across conditions. Signal weighting is also lower under Experience by 0.138 (SE 0.072, $p = 0.056$). The secondary experienced-frequency specification yields the same signs, with a somewhat larger and conventionally significant signal difference, but uses a different regressor and selected Experience subsample. These results therefore provide suggestive evidence of lower signal weighting under Experience, rather than a precise estimate of an aggregate description-experience gap. One possible interpretation is that learning an unknown base rate through sampling places additional demands on attention; the present design does not identify that mechanism.

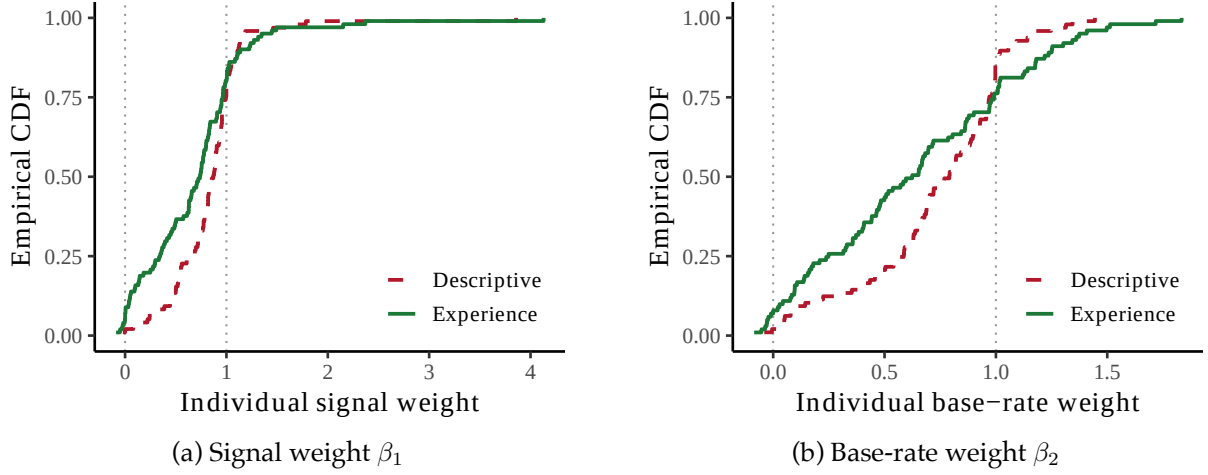


Figure 5: Individual signal and base-rate weights by condition. *Note:* Each panel plots the empirical cumulative distribution of subject-level OLS weights from the Grether regression (8), for the Description (dashed) and Experience (solid) conditions. Dotted vertical lines mark a weight of 0 (the cue is ignored) and 1 (Bayesian weighting). Kolmogorov–Smirnov tests reject equal distributions for both weights (β_1 : $p = 0.01$; β_2 : $p < 0.01$). Panel (a): median $\beta_1 = 0.87$ under Description versus 0.74 under Experience; Panel (b): median $\beta_2 = 0.79$ versus 0.62.

These aggregate estimates conceal substantial heterogeneity across subjects. To expose it, I re-estimate the Grether regression (Equation 8) subject by subject, recovering an individual signal weight β_1 and base-rate weight β_2 , and compare their empirical cumulative distributions across conditions in Figure 5. Kolmogorov–Smirnov tests reject equal distributions for both weights (β_1 : $p = 0.01$; β_2 : $p < 0.01$). For the signal weight (Panel a), the distribution under Description lies almost everywhere to the right of that under Experience — subjects under Description weight the signal more heavily at nearly every quantile — consistent with the aggregate pattern.

Result 5. *The distribution of signal weights is lower under Experience than under Description, consistent with lower aggregate signal weighting.*

For the base-rate weight (Panel b), a richer picture emerges. Across most of the distribution, subjects under Description place more weight on the base rate than their counterparts under Experience; yet the Experience condition also carries a heavier upper tail, with close to a quarter of its subjects (24%, 24/101) overweighting the base rate ($\beta_2 > 1$), against 12% (12/97) under Description (Fisher’s exact test, $p = 0.04$). Subjects who overweight the base rate draw substantially more cards than the rest, a mean of 17.7 versus 10.7 draws per round (permutation $p < 0.001$), and across all subjects under Experience the individual base-rate weight rises with sampling effort (Spearman

$\rho = 0.44, p < 0.001$). These associations are consistent with intensive sampling contributing to the upper tail, but do not establish that mechanism.

Result 6. *Aggregate base-rate weighting is imprecisely estimated across conditions, while the distributions of individual base-rate weights differ and the Experience upper tail is consistent with heterogeneity in sampling intensity.*

5 Discussion

This paper develops a cognitive account of base-rate neglect (BRN) across Description and Experience. The experiment yields three main findings. First, subjects substantially underweight the base rate under both paradigms, whether the experienced base rate is evaluated against the objective deck composition or against the subject’s own experienced frequency. Second, subjects with sharper numerical acuity place greater weight on the base rate in both conditions. Third, Experience introduces an information-acquisition stage: subjects draw fewer cards at more extreme base rates, and subjects who draw more cards on average place greater weight on the base rate. The comparison across paradigms does not identify a precise difference in average base-rate weighting. Its clearer contribution is distributional: base-rate weights are more heterogeneous under Experience, with heavier lower and upper tails.

A cognitive foundation for BRN. The framework provides a cognitive micro-foundation for BRN. Rather than treating the base-rate coefficient in a reduced-form updating rule as a fixed departure from Bayes’ rule, it derives the coefficient from the precision with which the base rate is encoded. That precision reflects both a basic capacity for numerical encoding and environmental features that determine exposure to base-rate information. BRN is therefore endogenous and context-dependent, rather than a stable individual characteristic. This interpretation places BRN within the noisy coding approach to economic behavior (Khaw, Li and Woodford, 2021; Enke and Graeber, 2023; Augenblick, Lazarus and Thaler, 2025; Oprea and Vieider, 2025). It also cautions against interpreting a base-rate weight as a direct measure of a subject’s general capacity for probabilistic reasoning without considering how the informational environment supports numerical representation (Ba, Bohren and Imas, 2024; Chan et al., 2025).

Experience is not just another format. The Experience paradigm should not be understood as simply another way of displaying the same information. It requires subjects to acquire the base rate before representing and using it. The resulting variation in sampling behavior offers a natural source of heterogeneity in base-rate weighting. Relative to Description, Experience contains both more subjects who assign very little weight to the base rate ($\beta_2 < 0.5$) and more who overweight it ($\beta_2 > 1$). The two conditions studied here are polar cases: the base rate is either stated in full or learned entirely through voluntary sampling. An informative intermediate case would state the base rate but also allow subjects to take additional samples. Such redundant sampling could sharpen a stated representation or introduce new finite-sample distortions, depending on how much information is sampled. In risky choice, [Oprea and Vieider \(2025\)](#) show that forcing redundant sampling reduces representational imprecision; whether the same intervention improves base-rate use is a direct question for future work.

Interactions among information channels. The experiment also finds weaker signal weighting under Experience, even though the experimental manipulation concerns only the base rate. This finding suggests that the acquisition and representation of base-rate information can affect the use of other information. [Chan et al. \(2025\)](#) provide a closely related result: longer exposure to an objectively given frequency grid raises the weight subjects place on the prior while lowering the weight they place on the signal. Together, the two studies suggest that the balance between prior and signal use responds to how prior information is represented. This possibility is consistent with work showing that multiple cognitive constraints can jointly shape updating. [Ba, Bohren and Imas \(2024\)](#) show how limited attention in representing a multistate environment interacts with cognitive imprecision in processing signals, while [Vieider \(2026\)](#) show that greater attention to a more difficult attribute can coexist with lower behavioral sensitivity to that attribute. Whether the weaker signal weighting here reflects attention reallocation, greater task complexity, or another processing channel is an important question for future research.

Policy implications. The model suggests possible ways to reduce BRN. On the one hand, structured exposure to a large, representative sample may sharpen the encoded base rate, as may interventions that improve numerical representation. These possibil-

ities can be evaluated by varying exposure time, sample size, or the visual representation of the base rate. For example, [Oprea and Vieider \(2025\)](#) show in risky choice that forcing subjects to observe redundant samples can reduce representational imprecision and eliminate the description–experience gap. A complementary possibility is training directed at the approximate number system. [Park and Brannon \(2013\)](#) find that such training improves mathematical proficiency, making it a promising intervention to evaluate for BRN.

A Extra Tables and Figures

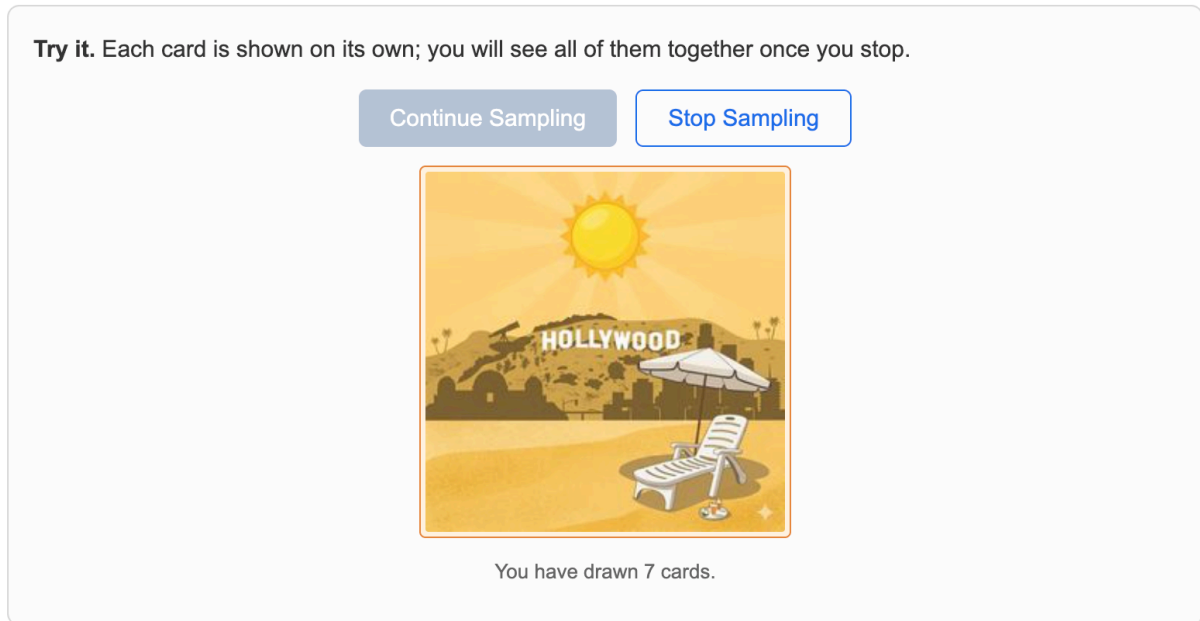


Figure A.1: Experiment screenshot of the voluntary-sampling interface, mid-draw.

Table A.1: Descriptive statistics across conditions.

	Description ($N = 97$)	Experience ($N = 101$)	Test	p
Female	48% (47/97)	45% (45/101)	χ^2	0.684
Age (years)	32 [29, 44]	35 [28, 42]	Wilcoxon	0.942
Education	5 [4, 5]	5 [4, 5]	Wilcoxon	0.561
Duration (min)	25.2 [19.4, 32.4]	26.8 [21.4, 35.3]	Wilcoxon	0.057

Note: $N = 198$ subjects (97 Description, 101 Experience). Cells report the median with the interquartile range in brackets, except *Female*, which reports the percentage (and count) of women. Duration is total completion time. No difference is significant at the 5% level; completion time is marginally longer under Experience ($p = 0.06$).

Table A.2 lists the full set of stimuli under Description and reports the mean reported posterior in each cell alongside the Bayesian benchmark; the aggregate pattern is shown in Figure 3a in the main text. Table A.5 reports the same quantities under Experience.

Table A.2: Stimuli and reported posteriors under Description.

		Sunny Day					Rainy Day				
		Lld.	Bayes	Reported			Lld.	Bayes	Reported		
Base rate				Avg.	S.E.	Devi.			Avg.	S.E.	Devi.
<i>Base-rate variation</i>											
1	0.05	0.7	0.109	0.228	0.025	-0.119	0.3	0.022	0.099	0.016	-0.077
2	0.1	0.7	0.206	0.372	0.026	-0.166	0.3	0.045	0.129	0.016	-0.084
3	0.2	0.7	0.368	0.429	0.020	-0.060	0.3	0.097	0.195	0.018	-0.099
4	0.3	0.7	0.500	0.511	0.017	-0.011	0.3	0.155	0.242	0.017	-0.087
5	0.4	0.7	0.609	0.585	0.015	0.024	0.3	0.222	0.285	0.015	-0.063
6	0.5	0.7	0.700	0.687	0.011	0.013	0.3	0.300	0.333	0.014	-0.033
7	0.6	0.7	0.778	0.719	0.013	0.059	0.3	0.391	0.429	0.015	-0.038
8	0.7	0.7	0.845	0.781	0.013	0.064	0.3	0.500	0.478	0.017	0.022
9	0.8	0.7	0.903	0.825	0.013	0.078	0.3	0.632	0.542	0.020	0.090
10	0.9	0.7	0.955	0.871	0.017	0.083	0.3	0.794	0.662	0.024	0.132
11	0.95	0.7	0.978	0.905	0.011	0.073	0.3	0.891	0.761	0.025	0.129
<i>Signal-structure variation</i>											
12	0.5	0.6	0.600	0.613	0.009	-0.013	0.4	0.400	0.394	0.008	0.006
13	0.5	0.8	0.800	0.745	0.016	0.055	0.2	0.200	0.286	0.020	-0.086
14	0.5	0.9	0.900	0.846	0.015	0.054	0.1	0.100	0.209	0.023	-0.109

Note: Description condition, $N = 97$ subjects per cell. Base rate is the share of Orange City cards in the deck; Lld. is the probability of the drawn day's weather in the Orange City; Bayes is the Bayesian posterior that the day comes from the Orange City. Reported gives the mean and standard error of subjects' reports; Devi. is Bayes minus the mean report, so a negative value means reports lie above the Bayesian posterior.

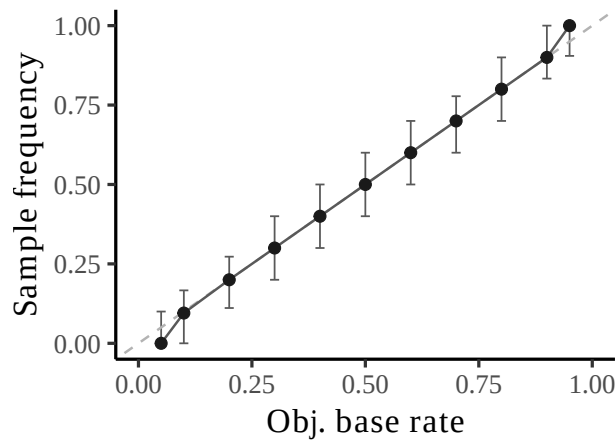


Figure A.2: Median sampled frequency by true base rate. *Note:* Experience condition, base-rate variation block (likelihood 0.7), retest round excluded. Each point is the median across rounds of the sampled frequency $\hat{p} = n_{\text{orange}}/n_{\text{draws}}$ at each true base rate; bars give the interquartile range, and the dashed line is the 45° line. Median counterpart to Figure 4c.

Table A.3: Dot-comparison accuracy as a stimulus-driven measure.

	Description			Experience		
	(1)	(2)	(3)	(4)	(5)	(6)
In larger ratio	3.989*** (0.297)	3.997*** (0.297)	5.198** (2.081)	4.128*** (0.293)	4.130*** (0.293)	6.063*** (2.350)
Female		-0.120 (0.104)	-0.017 (0.163)		0.016 (0.088)	-0.276* (0.157)
Age		-0.002 (0.005)	-0.012 (0.009)		-0.000 (0.004)	-0.003 (0.007)
Education		-0.077 (0.070)	0.063 (0.126)		-0.082 (0.062)	0.063 (0.106)
CRT score		0.075 (0.058)	0.016 (0.089)		0.014 (0.037)	-0.017 (0.063)
In ratio $\times$ Female			-0.449 (0.561)			1.275* (0.652)
In ratio $\times$ Age			0.044 (0.035)			0.012 (0.033)
In ratio $\times$ Education			-0.606 (0.425)			-0.628 (0.468)
In ratio $\times$ CRT score			0.255 (0.314)			0.138 (0.255)
Constant	0.113 (0.082)	0.473 (0.441)	0.199 (0.653)	0.022 (0.072)	0.394 (0.363)	-0.053 (0.578)
Trials	3880	3880	3880	4040	4040	4040
Subjects	97	97	97	101	101	101

Note: Trial-level logistic regressions of a correct dot-comparison choice on the log of the larger ratio of the two sets, $\ln[\max(n_{\text{yellow}}, n_{\text{blue}}) / \min(n_{\text{yellow}}, n_{\text{blue}})]$, and subject characteristics, estimated separately for the Description (columns 1–3) and Experience (columns 4–6) conditions. Columns (1) and (4) are the baseline; columns (2) and (5) add the characteristics; columns (3) and (6) add their interactions with the log ratio. Standard errors, clustered by subject, in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

Table A.4: Grether regression under Description: full specification.

	(1)	(2)	(3)	(4)
β_0 (constant)	0.028 (0.017)	-0.089 (0.079)	0.028 (0.017)	0.028 (0.017)
β_1 (signal, ℓ_S)	0.840*** (0.044)	0.735*** (0.061)	0.840*** (0.044)	0.840*** (0.044)
β_2 (base rate, ℓ)	0.728*** (0.033)	0.729*** (0.036)	0.728*** (0.032)	0.729*** (0.031)
$\ell \times$ sunny-day signal		-0.003 (0.025)		
$\ell_S \times$ numerical acuity			-0.002 (0.054)	-0.002 (0.054)
$\ell \times$ numerical acuity			0.104*** (0.024)	0.096*** (0.024)
$\ell \times$ CRT				0.013 (0.034)
$\ell \times$ page time (between)				0.069** (0.032)
$\ell \times$ page time (within)				-0.010 (0.011)
Sunny-day signal		0.235 (0.161)		
Numerical acuity			0.003 (0.024)	0.009 (0.024)
CRT				-0.018 (0.018)
Page time (between)				-0.013 (0.015)
Page time (within)				0.009 (0.019)
Subjects	97	97	97	97
Observations	2716	2716	2716	2716

Note: Description condition, retest round excluded; the dependent variable is ℓ_{Post} . Column (1) is the baseline Grether regression (8); column (2) interacts the prior with the signal realization; columns (3) and (4) reproduce columns (2) and (3) of Table 1, adding the interaction of the signal with numerical acuity and the main effects of all moderators. Standard errors, clustered by subject, in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

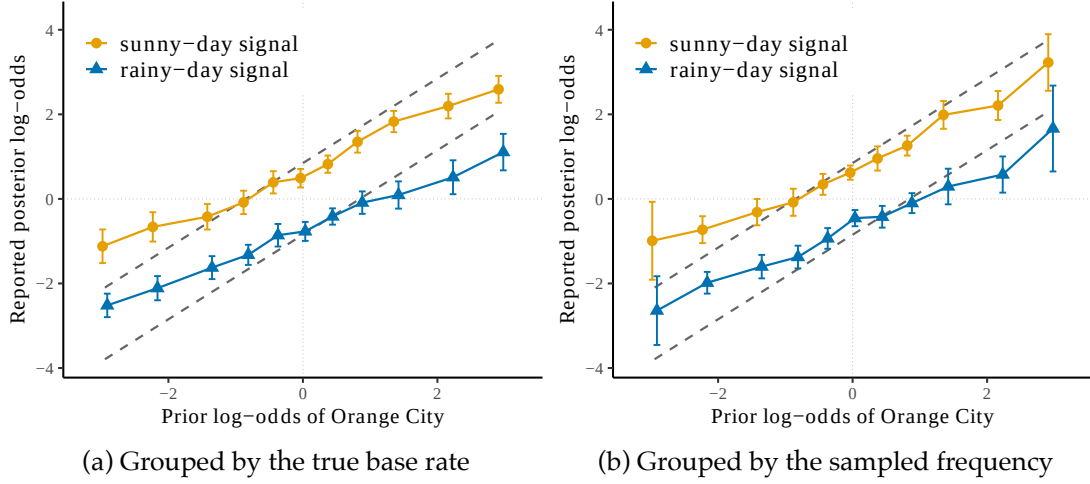


Figure A.3: Belief updating under Experience: true base rate versus sampled frequency. *Note:* Experience condition, base-rate variation block (likelihood 0.7), retest round excluded. Each panel plots the mean reported posterior log-odds ℓ_{Post} against prior log-odds: in panel (a) the prior is the true base rate; in panel (b) it is the subject's sampled frequency $\hat{p} = n_{orange}/n_{draws}$, using only rounds with at least one card of each city and rounded to the nearest design level. Orange circles: sunny-day signal; blue triangles: rainy-day signal. Dashed lines are the Bayesian posteriors. Error bars are 95% confidence intervals across subjects. Both panels share the same axes.

Table A.5: Stimuli and reported posteriors under Experience.

		Sunny Day					Rainy Day				
		Lld.	Bayes	Reported			Lld.	Bayes	Reported		
Base rate	Avg.			S.E.	Devi.	Avg.			S.E.	Devi.	
<i>Base-rate variation</i>											
1	0.05	0.7	0.109	0.329	0.030	-0.220	0.3	0.022	0.132	0.014	-0.110
2	0.1	0.7	0.206	0.392	0.028	-0.186	0.3	0.045	0.166	0.017	-0.121
3	0.2	0.7	0.368	0.435	0.026	-0.066	0.3	0.097	0.227	0.019	-0.130
4	0.3	0.7	0.500	0.486	0.023	0.014	0.3	0.155	0.259	0.018	-0.104
5	0.4	0.7	0.609	0.579	0.023	0.029	0.3	0.222	0.347	0.021	-0.125
6	0.5	0.7	0.700	0.609	0.020	0.091	0.3	0.300	0.356	0.019	-0.056
7	0.6	0.7	0.778	0.662	0.018	0.115	0.3	0.391	0.415	0.019	-0.023
8	0.7	0.7	0.845	0.736	0.019	0.109	0.3	0.500	0.485	0.023	0.015
9	0.8	0.7	0.903	0.806	0.016	0.097	0.3	0.632	0.520	0.026	0.112
10	0.9	0.7	0.955	0.836	0.018	0.118	0.3	0.794	0.576	0.030	0.218
11	0.95	0.7	0.978	0.859	0.018	0.119	0.3	0.891	0.653	0.032	0.238
<i>Signal-structure variation</i>											
12	0.5	0.6	0.600	0.609	0.019	-0.009	0.4	0.400	0.404	0.020	-0.004
13	0.5	0.8	0.800	0.654	0.020	0.146	0.2	0.200	0.364	0.022	-0.164
14	0.5	0.9	0.900	0.701	0.024	0.199	0.1	0.100	0.294	0.022	-0.194

Note: Experience condition, $N = 101$ subjects per cell. Columns are defined as in Table A.2. Subjects learn the deck by drawing cards rather than being told its composition; the Bayesian benchmark uses the true base rate.

Table A.6: Grether regression under Experience: prior measure versus sample.

	A: Obj. base rate All rounds		B: Sampled frequency Interior rounds		C: Obj. base rate Interior rounds	
	(1)	(2)	(3)	(4)	(5)	(6)
β_0 (constant)	-0.019 (0.029)	-0.019 (0.029)	0.000 (0.030)	-0.002 (0.030)	-0.011 (0.032)	-0.012 (0.032)
β_1 (signal, ℓ_S)	0.702*** (0.057)	0.702*** (0.057)	0.687*** (0.060)	0.687*** (0.060)	0.687*** (0.060)	0.687*** (0.060)
β_2 (base rate, ℓ)	0.641*** (0.046)	0.638*** (0.041)	0.679*** (0.051)	0.638*** (0.048)	0.563*** (0.049)	0.520*** (0.043)
$\ell \times$ numerical acuity		0.093*** (0.026)		0.071** (0.028)		0.068*** (0.026)
$\ell \times$ CRT		-0.026 (0.046)		0.006 (0.057)		-0.018 (0.052)
$\ell \times$ page time (between)		-0.048 (0.051)		-0.064 (0.053)		-0.021 (0.048)
$\ell \times$ page time (within)		-0.012 (0.014)		0.006 (0.024)		-0.012 (0.016)
$\ell \times$ draws (between)		0.187*** (0.048)		0.131** (0.051)		0.185*** (0.046)
$\ell \times$ draws rank (within)		-0.027 (0.031)		-0.049 (0.031)		0.000 (0.032)
Subjects	101	101	101	101	101	101
Observations	2828	2828	2322	2322	2322	2322

Note: Experience condition, retest round excluded. Panels A and B reproduce columns (1), (3), (4) and (6) of Table 2. Panel C uses the objective base rate as the prior on the same interior rounds as Panel B (rounds with at least one card of each city), so Panels B and C differ only in the prior and Panels A and C only in the sample. Odd columns are the baseline Grether regression (8); even columns add the interactions of the prior with numerical acuity, CRT, between- and within-subject page time, and the subject's mean draws per round and within-subject draw rank. All moderators are standardized. Standard errors, clustered by subject, in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

B Beta-Binomial Moments and the Grether Mapping

This appendix records the posterior moments of the Beta-Binomial decoding step and maps the model's average decoded belief into the Grether regression (8). The key point is that the model is linear in probability space but nonlinear in log-odds, so the Grether coefficients summarize a curved mapping away from the center of the probability scale.

Exact posterior moments. Let the DM's prior over the state probability p be $\text{Beta}(a_0, b_0)$ and let $\hat{a} \sim \text{Binomial}(\kappa, p)$ be the encoded count, with $\hat{b} = \kappa - \hat{a}$. Conjugacy gives

$$p \mid \hat{a} \sim \text{Beta}(a_0 + \hat{a}, b_0 + \hat{b}), \quad (\text{B1})$$

and therefore

$$\mathbb{E}[p \mid \hat{a}] = \frac{a_0 + \hat{a}}{a_0 + b_0 + \kappa}, \quad \text{Var}[p \mid \hat{a}] = \frac{(a_0 + \hat{a})(b_0 + \hat{b})}{(a_0 + b_0 + \kappa)^2 (a_0 + b_0 + \kappa + 1)}, \quad (\text{B2})$$

with $(\alpha, \beta) = (a_0 + \hat{a}, b_0 + \hat{b})$. The mean is the realized decoded base rate whose expectation gives (5); the variance records the uncertainty surrounding that realization. Both are well-defined even at the corners $\hat{a} = 0$ and $\hat{a} = \kappa$ because the default counts are positive.

The exact log-odds map. Under the neutral ignorance prior $p_0 = \frac{1}{2}$, averaging over internal-sampling noise gives $p_{BR} = \gamma p + (1 - \gamma)\frac{1}{2}$. Substituting $p = e^\ell / (1 + e^\ell)$ yields

$$\ell_{BR} = \ln \frac{(1 - \gamma) + (1 + \gamma)e^\ell}{(1 + \gamma) + (1 - \gamma)e^\ell}. \quad (\text{B3})$$

This map is odd, strictly increasing, and nonlinear. It passes through the origin, but its slope varies with ℓ , so (8) is not an exact identity in the objective log-odds; its coefficients summarize the best straight-line fit to (B3).

The slope and β_2 . Let $g(x) = \ln(x/(1 - x))$, so that $\ell = g(p)$ and $\ell_{BR} = g(p_{BR})$. Since $dp_{BR}/dp = \gamma$, the local log-odds slope is

$$\frac{d\ell_{BR}}{d\ell} = \frac{g'(p_{BR})(dp_{BR}/dp)}{g'(p)} = \gamma \cdot \frac{p(1 - p)}{p_{BR}(1 - p_{BR})}. \quad (\text{B4})$$

Writing $\delta = p - \frac{1}{2}$, the neutral-prior case gives $p(1 - p) = \frac{1}{4} - \delta^2$ and $p_{BR}(1 - p_{BR}) = \frac{1}{4} - \gamma^2\delta^2$, hence

$$\frac{d\ell_{BR}}{d\ell} = \gamma \frac{1 - 4\delta^2}{1 - 4\gamma^2\delta^2}, \quad (\text{B5})$$

which decreases in $|\delta|$ for $\gamma < 1$. At the neutral base rate,

$$\left. \frac{d\ell_{BR}}{d\ell} \right|_{p=1/2} = \gamma, \quad (\text{B6})$$

so the structural evidence weight and the local log-odds slope coincide exactly at the center.

This is the sense in which β_2 identifies γ . Locally around $p = \frac{1}{2}$, the two are equal. In the symmetric experimental design, a Grether regression fitted over a wider range of base rates averages the smaller local slopes away from the center, so β_2 tracks γ but lies below it whenever non-central base rates receive positive weight.

The intercept under a neutral prior. The same symmetry pins down the central intercept. With $p_0 = \frac{1}{2}$, the decoded base rate at $p = \frac{1}{2}$ is still exactly $\frac{1}{2}$, so

$$\ell_{BR}|_{p=1/2} = \ln \frac{1/2}{1/2} = 0. \quad (\text{B7})$$

Encoding noise then compresses base rates toward even odds without tilting them toward either state. Because (B3) is odd, any least-squares Grether regression on a base-rate design symmetric around $\ell = 0$ also has fitted intercept $\beta_0 = 0$. With an asymmetric design, the fitted intercept can absorb curvature, but the structural central intercept remains zero.

The intercept under a non-neutral prior. If $p_0 \neq \frac{1}{2}$, the prior breaks this symmetry. The central intercept is

$$\beta_0 = \ell_{BR}|_{p=1/2} = \ln \frac{p_{BR}(0)}{1 - p_{BR}(0)}, \quad p_{BR}(0) = \frac{\gamma}{2} + (1 - \gamma)p_0.$$

Letting $\delta_0 = p_0 - \frac{1}{2}$,

$$2p_{BR}(0) = \gamma + 2(1 - \gamma)p_0 = \gamma + (1 - \gamma)(1 + 2\delta_0) = 1 + 2(1 - \gamma)\delta_0, \quad (\text{B8})$$

and $2(1 - p_{BR}(0)) = 1 - 2(1 - \gamma)\delta_0$. Thus

$$\beta_0 = \ln \frac{1 + 2(1 - \gamma)\delta_0}{1 - 2(1 - \gamma)\delta_0} = 2 \operatorname{artanh}[(1 - \gamma)(2p_0 - 1)], \quad \delta_0 = p_0 - \frac{1}{2}. \quad (\text{B9})$$

Equivalently, with $d \equiv (1 - \gamma)(2p_0 - 1)$, $\beta_0 = \ln[(1 + d)/(1 - d)]$ and the central slope is $\beta_2 = \gamma/(1 - d^2)$. The neutral case sets $d = 0$, giving $\beta_0 = 0$ and $\beta_2 = \gamma$; for small displacements, $\beta_0 \approx 4(1 - \gamma)\delta_0 \approx (1 - \gamma)\ell_0$. Equation (B9) is a *central* intercept, not a design-free formula for the fitted OLS intercept: with a non-neutral prior, the full map

is asymmetric, so the projected intercept depends on the distribution of base rates. The central sign result is nevertheless exact: if $\gamma < 1$, then β_0 has the sign of $p_0 - \frac{1}{2}$.

C The One-Step-Lookahead Sampling Rule

The main text uses a one-step-lookahead rule: the DM takes another draw when its expected reduction in the dispersion of the running frequency exceeds the subjective cost c . This follows the myopic information-acquisition logic of [Vieider \(2026\)](#), adapted here to a single unknown base rate and a belief report rather than a choice between options.

Let $q = \hat{p}_s$ denote the running frequency after s draws. Using the current frequency as the predictive probability for the next draw, let $X \sim \text{Bernoulli}(q)$ and $\hat{p}_{s+1} = (sq + X)/(s + 1)$. The expected dispersion after one more draw is

$$\mathbb{E}[\hat{V}_{s+1} \mid \hat{p}_s = q] = \mathbb{E}\left[\frac{\hat{p}_{s+1}(1 - \hat{p}_{s+1})}{s + 1} \mid q\right] = q(1 - q)\frac{s(s + 2)}{(s + 1)^3}. \quad (\text{C1})$$

The exact expected reduction from one additional draw is therefore

$$\Delta V_s(q) = q(1 - q) \left[\frac{1}{s} - \frac{s(s + 2)}{(s + 1)^3} \right] = q(1 - q) \frac{s^2 + 3s + 1}{s(s + 1)^3} = \frac{q(1 - q)}{s^2} + O(s^{-4}). \quad (\text{C2})$$

The leading term gives the continuation condition used in the main text, $q(1 - q)/s^2 \geq c$, and the approximate target sample size in [\(11\)](#). The rule is not a solution to a full dynamic program: it compares one additional draw with stopping immediately.

The rule also clarifies the predicted shape across base rates. A sample path generated under p has a mirror path under $1 - p$ obtained by replacing every draw with its opposite; because $q(1 - q)$ is unchanged, the stopping-time distribution is symmetric around $p = 1/2$. Moreover, running frequencies tend to be more lopsided when the true base rate is more extreme, lowering [\(C2\)](#). Expected sample size should therefore be largest near one-half and smaller toward either extreme, as stated in [Prediction 2](#).

D The Dot-Comparison Task

To obtain a measure of numerical acuity outside the belief task, participants complete a nonsymbolic numerical discrimination task modeled on [Halberda, Mazzocco and](#)

Feigenson (2008). The task measures the acuity of the approximate number system (ANS) — the precision with which a person represents the approximate numerosity of a set without verbal counting — which Halberda, Mazocco and Feigenson (2008) show varies substantially across individuals and is well summarized by a single subject-level parameter.

Procedure. On each trial the subject sees, after a 250 ms fixation, a single array of spatially intermixed blue and yellow dots presented for 300 ms — too briefly to count serially — and judges by keypress which color is more numerous. Each set contains between 5 and 16 dots, and the ratio of the smaller to the larger set is drawn at random from four bins, 1 : 2, 3 : 4, 5 : 6, and 7 : 8, spanning easy to hard discriminations; which color is more numerous is randomized across trials. Following Halberda, Mazocco and Feigenson (2008), half of the trials at each ratio are *dot-size controlled* (the average blue and yellow dots are equal in size, so the more numerous set also covers the larger area) and half are *area controlled* (the two colors cover equal total area, so the more numerous set has smaller dots on average), and individual dot sizes are jittered from trial to trial. Together these controls reduce the scope for judgments based on cumulative area, density, or dot size rather than numerosity. Subjects first complete a short block of practice trials, then a block of 40 test trials (10 per ratio bin); correct-or-incorrect feedback is shown after each trial in both blocks.¹⁴

Estimating acuity. Each subject’s performance is summarized by a single Weber fraction w , the noise in the internal representation of numerosity relative to the represented magnitude. Under the scalar-variability account of the ANS, numerosities are encoded on a logarithmic internal scale, so a trial contrasting counts $n_1 < n_2$ is discriminated according to the log ratio of the two magnitudes, and the probability of a correct response is

$$\Pr(\text{correct}) = \Phi\left(\frac{\ln(n_2/n_1)}{\sqrt{2} w}\right), \quad (\text{D1})$$

where $\Phi(\cdot)$ is the standard normal distribution function. We estimate each subject’s $\hat{w}$ by maximum likelihood over that subject’s trials, treating each judgment as an independent Bernoulli outcome with the success probability in (D1). A smaller $\hat{w}$ indicates

¹⁴In the descriptive condition, a second 40-trial test block followed a short break. The acuity estimate below uses the first block only, so it rests on the same 40 trials for every subject.

sharper acuity (finer numerical discrimination), a larger $\hat{w}$ coarser acuity. For reference, Halberda, Mazocco and Feigenson (2008) report adult Weber fractions ranging from about 0.12 to 0.57.¹⁵

Mapping the Weber fraction to encoding precision. Under the maintained assumption that the dot task and the base-rate tasks draw on a common magnitude code, the Weber fraction can be related locally to the encoding precision of (3). Let $p = n_2/(n_1 + n_2) > 1/2$ denote the share of dots belonging to the more numerous color and let $\kappa_{\text{dot}} = \kappa(e_{\text{dot}}; \theta)$ denote the precision with which that share is encoded. Because every subject views each array for the same 300 ms under the same display protocol, the task holds important components of exposure fixed. Cross-subject variation in $\hat{w}$ can therefore serve as an inverse proxy for θ under the common-code assumption, although attention, perceptual noise, and response error may also contribute to measured performance. To make the directional link explicit, suppose the share is represented by κ_{dot} internal samples as in (2). The subject chooses the more numerous color when the encoded share exceeds one half, so the normal approximation to the Binomial gives

$$\Pr(\text{correct}) = \Phi\left(\frac{\sqrt{\kappa_{\text{dot}}}\left(p - \frac{1}{2}\right)}{\sqrt{p(1-p)}}\right), \quad (\text{D2})$$

Around $p = 1/2$, this is approximately $\Phi(2\sqrt{\kappa_{\text{dot}}}(p - 1/2))$. Since $\ln(n_2/n_1) = \ln(p/(1-p)) \approx 4(p - 1/2)$ near one half, the Weber model (D1) is locally $\Phi(2\sqrt{2}(p - 1/2)/w)$. Matching the two local slopes gives

$$\kappa_{\text{dot}} \approx \frac{2}{w^2}. \quad (\text{D3})$$

This correspondence is local and establishes an ordering rather than a literal calibration across tasks: a lower w maps to a higher κ_{dot} . A symmetric default does not affect that ordering because shrinking the encoded share toward one half does not move it across the comparison threshold. Since exposure and presentation differ between the dot and belief tasks, $2/\hat{w}^2$ does not identify the concentration used in either belief task.

¹⁵As a robustness check we also compute an alternative Weber fraction from the linear-difference model of Halberda, Mazocco and Feigenson (2008), $\Pr(\text{correct}) = 1 - \frac{1}{2} \operatorname{erfc}((n_2 - n_1)/(\sqrt{2}w\sqrt{n_1^2 + n_2^2}))$, in which the two sets are Gaussian with means n_i and standard deviations $w n_i$. Fit by maximum likelihood on the same trials, it correlates with the log-ratio estimate at $r \approx 0.99$ across subjects, so the choice of functional form does not affect the results.

Use. I use the estimated acuity $\hat{w}$ as an inverse proxy for the magnitude-code precision θ in the model. Under the maintained common-code assumption, a lower $\hat{w}$ indicates a sharper magnitude code and therefore predicts higher encoding precision at a given level of exposure in both paradigms: $\kappa_D = \kappa(e; \theta)$ under description and $\kappa_E = \kappa(s; \theta)$ under experience. I test Prediction 1 and Prediction 3a by interacting the standardized negative Weber fraction with prior log-odds in the Grether regression, separately by condition; the predicted interaction is positive in each because sharper acuity should be associated with weaker neglect.

References

- Abdellaoui, Mohammed, Olivier L'Haridon, and Corina Paraschiv.** 2011. "Experienced vs. Described Uncertainty: Do We Need Two Prospect Theory Specifications?" *Management Science*, 57(10): 1879–1895.
- Araujo, Felipe A, Christopher F Chabris, and Michelle N Meyer.** 2025. "The Effect of On-the-Job Experience on Base-Rate Neglect: Evidence from Medical Professionals." *Management Science*, 71(10): 8807–8822.
- Augenblick, Ned, Eben Lazarus, and Michael Thaler.** 2025. "Overinference from Weak Signals and Underinference from Strong Signals." *The Quarterly Journal of Economics*, 140(1): 335–401.
- Ba, Cuimin, J. Aislinn Bohren, and Alex Imas.** 2024. "Over- and Underreaction to Information." Working paper.
- Barbey, Aron K., and Steven A. Sloman.** 2007. "Base-Rate Respect: From Ecological Rationality to Dual Processes." *Behavioral and Brain Sciences*, 30(3): 241–254.
- Bar-Hillel, Maya.** 1980. "The Base-Rate Fallacy in Probability Judgments." *Acta Psychologica*, 44(3): 211–233.
- Barron, Greg, and Ido Erev.** 2003. "Small Feedback-Based Decisions and Their Limited Correspondence to Description-Based Decisions." *Journal of Behavioral Decision Making*, 16(3): 215–233.
- Benjamin, Daniel J.** 2019. "Errors in Probabilistic Reasoning and Judgment Biases." In *Handbook of Behavioral Economics: Applications and Foundations 1*. Vol. 2, , ed. B. Douglas Bernheim, Stefano DellaVigna and David Laibson. Elsevier.
- Bouchouicha, Ranoua, Ryan Oprea, Ferdinand M Vieider, and Jilong Wu.** 2025. "Cognitive Frictions and Canonical Patterns in Risk-Taking." Working paper.
- Brandts, Jordi, and Gary Charness.** 2011. "The strategy versus the direct-response method: a first survey of experimental comparisons." *Experimental Economics*, 14(3): 375–398.

- Casscells, Ward, Arno Schoenberger, and Thomas B. Graboys.** 1978. "Interpretation by Physicians of Clinical Laboratory Results." *New England Journal of Medicine*, 299(18): 999–1001.
- Chan, Kenneth, Gary Charness, Chetan Dave, and J. Lucas Reddinger.** 2025. "On Prior Confidence and Belief Updating." Working paper.
- Costello, Fintan, and Paul Watts.** 2014. "Surprisingly Rational: Probability Theory Plus Noise Explains Biases in Judgment." *Psychological Review*, 121(3): 463–480.
- Danz, David, Lise Vesterlund, and Alistair J. Wilson.** 2022. "Belief Elicitation and Behavioral Incentive Compatibility." *American Economic Review*, 112(9): 2851–2883.
- Dehaene, Stanislas.** 2003. "The neural basis of the Weber–Fechner law: a logarithmic mental number line." *Trends in cognitive sciences*, 7(4): 145–147.
- Eddy, David M.** 1982. "Probabilistic Reasoning in Clinical Medicine: Problems and Opportunities." In *Judgment under Uncertainty: Heuristics and Biases.*, ed. Daniel Kahneman, Paul Slovic and Amos Tversky, 249–267. Cambridge University Press.
- Enke, Benjamin, and Thomas Graeber.** 2023. "Cognitive Uncertainty." *The Quarterly Journal of Economics*, 138(4): 2021–2067.
- Enke, Benjamin, Thomas Graeber, and Ryan Oprea.** 2023. "Confidence, self-selection, and bias in the aggregate." *American Economic Review*, 113(7): 1933–1966.
- Esponda, Ignacio, Emanuel Vespa, and Sevgi Yuksel.** 2024. "Mental Models and Learning: The Case of Base-Rate Neglect." *American Economic Review*, 114(3): 752–782.
- Frydman, Cary, and Lawrence J. Jin.** 2022. "Efficient Coding and Risky Choice." *The Quarterly Journal of Economics*, 137(1): 161–213.
- Frydman, Cary, and Lawrence J. Jin.** 2025. "On the Source and Instability of Probability Weighting." Working Paper.
- Gigerenzer, Gerd, and Ulrich Hoffrage.** 1995. "How to improve Bayesian reasoning without instruction: frequency formats." *Psychological review*, 102(4): 684.

- Grether, David M.** 1980. "Bayes Rule as a Descriptive Model: The Representativeness Heuristic." *The Quarterly Journal of Economics*, 95(3): 537–557.
- Halberda, Justin, and Lisa Feigenson.** 2008. "Developmental Change in the Acuity of the "Number Sense": The Approximate Number System in 3-, 4-, 5-, and 6-Year-Olds and Adults." *Developmental Psychology*, 44(5): 1457–1465.
- Halberda, Justin, Michèle M. M. Mazzocco, and Lisa Feigenson.** 2008. "Individual Differences in Non-Verbal Number Acuity Correlate with Maths Achievement." *Nature*, 455(7213): 665–668.
- Hertwig, Ralph, Greg Barron, Elke U. Weber, and Ido Erev.** 2004. "Decisions from Experience and the Effect of Rare Events in Risky Choice." *Psychological Science*, 15(8): 534–539.
- Juslin, Peter, Håkan Nilsson, Anders Winman, and Marcus Lindskog.** 2011. "Reducing Cognitive Biases in Probabilistic Reasoning by the Use of Logarithm Formats." *Cognition*, 120(2): 248–267.
- Kahneman, Daniel, and Amos Tversky.** 1973. "On the Psychology of Prediction." *Psychological Review*, 80(4): 237–251.
- Khaw, Mel Win, Ziang Li, and Michael Woodford.** 2021. "Cognitive Imprecision and Small-Stakes Risk Aversion." *The Review of Economic Studies*, 88(4): 1979–2013.
- Koehler, Jonathan J.** 1996. "The Base Rate Fallacy Reconsidered: Descriptive, Normative, and Methodological Challenges." *Behavioral and Brain Sciences*, 19(1): 1–17.
- McDowell, Michelle, and Perke Jacobs.** 2017. "Meta-analysis of the effect of natural frequencies on Bayesian reasoning." *Psychological bulletin*, 143(12): 1273.
- Oprea, Ryan, and Ferdinand M. Vieider.** 2025. "Minding the Gap: On the Origins of Probability Weighting and the Description-Experience Gap." Working paper.
- Park, Joonkoo, and Elizabeth M. Brannon.** 2013. "Training the Approximate Number System Improves Math Proficiency." *Psychological Science*, 24(10): 2013–2019.
- Phillips, Lawrence D., and Ward Edwards.** 1966. "Conservatism in a Simple Probability Inference Task." *Journal of Experimental Psychology*, 72(3): 346–354.

- Piazza, Manuela, Philippe Pinel, Denis Le Bihan, and Stanislas Dehaene.** 2007. "A magnitude code common to numerosities and number symbols in human intraparietal cortex." *Neuron*, 53(2): 293–305.
- Piazza, Manuela, Véronique Izard, Philippe Pinel, Denis Le Bihan, and Stanislas Dehaene.** 2004. "Tuning curves for approximate numerosity in the human intraparietal sulcus." *Neuron*, 44(3): 547–555.
- Sanborn, Adam N., and Nick Chater.** 2016. "Bayesian Brains without Probabilities." *Trends in Cognitive Sciences*, 20(12): 883–893.
- Vieider, Ferdinand M.** 2024. "Decisions Under Uncertainty as Bayesian Inference on Choice Options." *Management Science*, 70(12): 9014–9030.
- Vieider, Ferdinand M.** 2026. "Sequential Sampling of Noisy Information: Attention, Stopping, and Risk-Taking." Working paper.
- Wulff, Dirk U., Max Mergenthaler-Canseco, and Ralph Hertwig.** 2018. "A Meta-Analytic Review of Two Modes of Learning and the Description-Experience Gap." *Psychological Bulletin*, 144(2): 140–176.
- Zhu, Jian-Qiao, Adam N. Sanborn, and Nick Chater.** 2020. "The Bayesian Sampler: Generic Bayesian Inference Causes Incoherence in Human Probability Judgments." *Psychological Review*, 127(5): 719–748.